Menus, Dialogs and UI Interaction

1. Develop an Android application implementing AppBar (Toolbar) and Options Menu.

Project Name: ToolbarMenuDemo

FileName: activity_main.xml

<?xml version="1.0" encoding="utf-8"?>

```
<LinearLayout

   xmlns:android="http://schemas.android.com/apk/res/android"
xmlns:app="http://schemas.android.com/apk/res-auto"

   android:id="@+id/main"

   android:layout_width="match_parent"

   android:layout_height="match_parent" android:orientation="vertical">

   <!-- Toolbar -->

   <androidx.appcompat.widget.Toolbar

       android:id="@+id/toolbar"

       android:layout_width="match_parent"

       android:layout_height="?attr/actionBarSize"

       android:background="?attr/colorPrimary"

       app:title="Toolbar Demo"

       app:titleTextColor="@android:color/white"/>

   <!--Main Content -->

   <TextView

       android:layout_width="wrap_content"

       android:layout_height="wrap_content"

       android:text="AppBar and Options Menu Example"
```

```
        android:textSize="22sp"

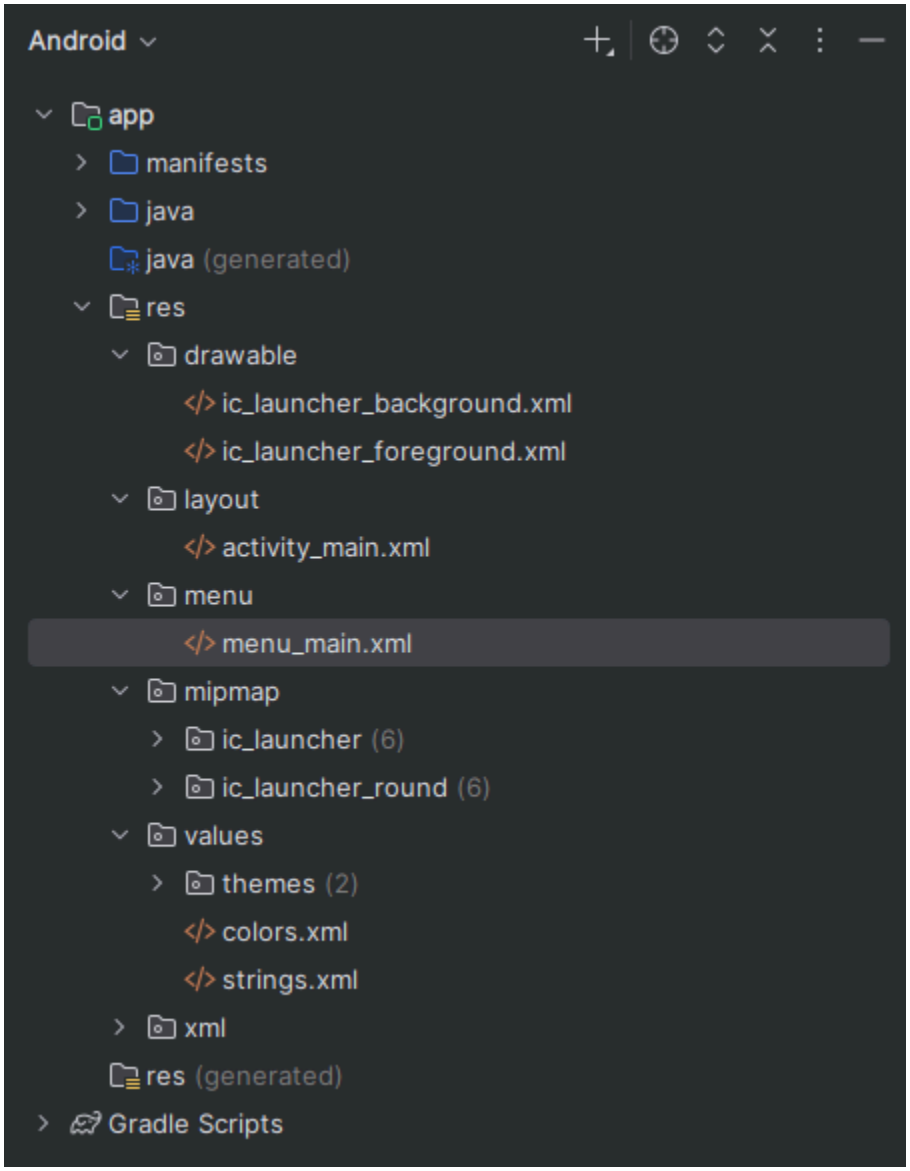
        android:padding="20dp"/>

</LinearLayout>
```

Create a Menu Resource File.
Right Click: Res → New→ Android Resource Directory

Directory Name: menu
Resource Type: menu

Create Menu XML File.
Right Click: menu New→ Menu Resource File Give name as **menu_main.xml**



File Name : menu_main.xml

```
<?xml version="1.0" encoding="utf-8"?>

<menu xmlns:android="http://schemas.android.com/apk/res/android"

   xmlns:app="http://schemas.android.com/apk/res-auto">



   <item

       android:id="@+id/menu_home"

       android:title="Home"

       app:showAsAction="never"/>



   <item

       android:id="@+id/menu_settings"

       android:title="Settings"

       app:showAsAction="never"/>

   <item

       android:id="@+id/menu_about"

       android:title="About"

       app:showAsAction="never"/>

   <item

       android:id="@+id/menu_logout"

       android:title="Logout"

       app:showAsAction="never"/>

</menu>
```

File Name: MainActivity.java.

```
package com.example.toolbarmenudemo;

import android.os.Bundle;
import android.view.Menu;
import android.view.MenuItem;
import android.widget.Toast;
import androidx.annotation.NonNull;
import androidx.appcompat.app.AppCompatActivity;
import androidx.appcompat.widget.Toolbar;

public class MainActivity extends AppCompatActivity {

   Toolbar toolbar;
   @Override
   protected void onCreate(Bundle savedInstanceState) {
       super.onCreate(savedInstanceState);
       setContentView(R.layout.activity_main);
       toolbar = findViewById(R.id.toolbar);
       // Set Toolbar as AppBar
       setSupportActionBar(toolbar);
   }
   // Create Options Menu
   @Override
   public boolean onCreateOptionsMenu(Menu menu) {
```

```
        getMenuInflater().inflate(R.menu.menu_main, menu);

        return true;

    }

    // Handle Menu Item Click

    @Override

    public boolean onOptionsItemSelected(@NonNull MenuItem item) {

        int id = item.getItemId();

        if (id == R.id.menu_home) {

            Toast.makeText(this, "Home Selected", Toast.LENGTH_SHORT).show();

        } else if (id == R.id.menu_settings) {

            Toast.makeText(this, "Settings Selected",
Toast.LENGTH_SHORT).show();

        } else if (id == R.id.menu_about) {

            Toast.makeText(this, "About Selected", Toast.LENGTH_SHORT).show();

        } else if (id == R.id.menu_logout) {

            Toast.makeText(this, "Logout Selected", Toast.LENGTH_SHORT).show();

        }

        return true;

    }

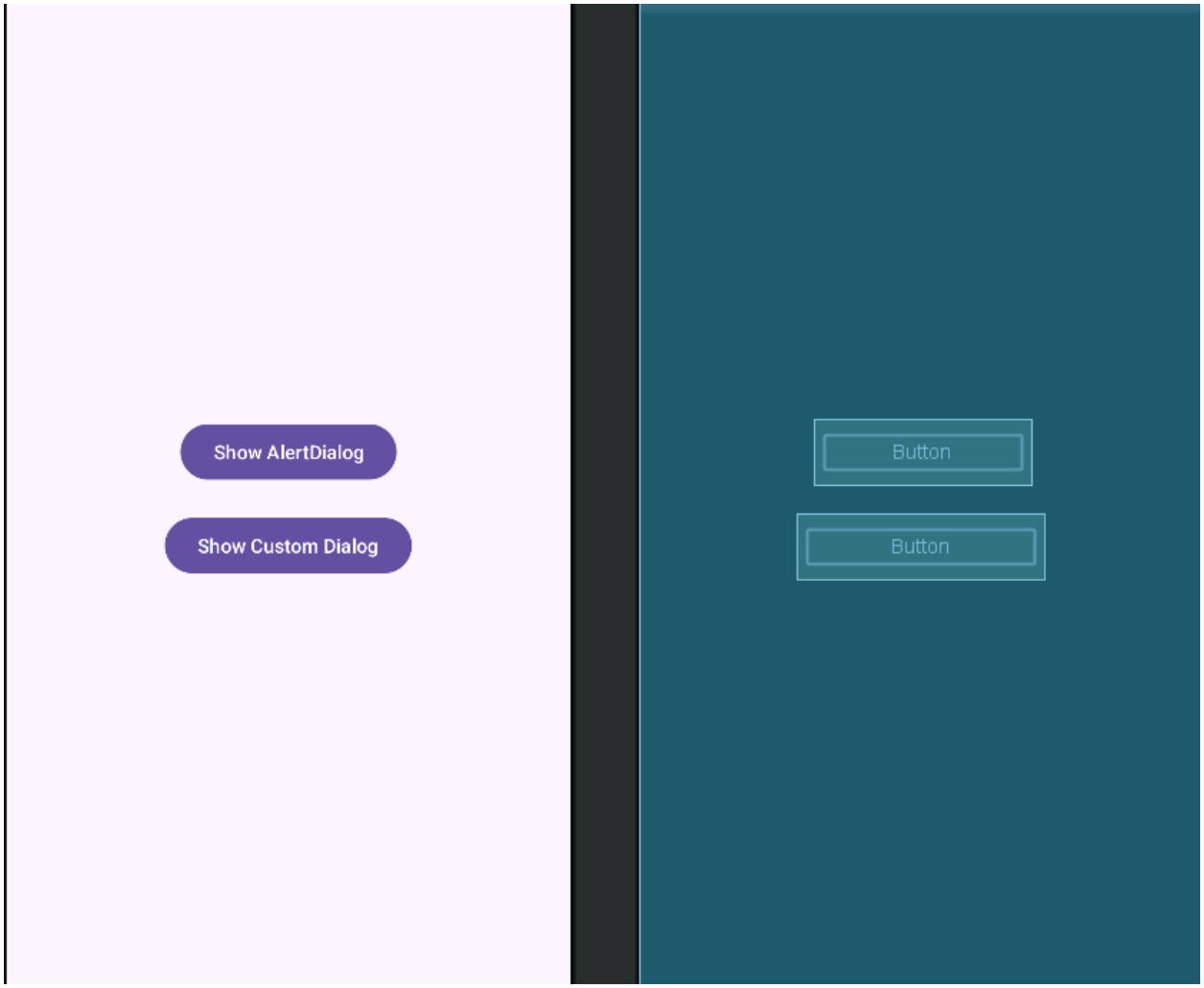
}
```


Toolbar Demo
AppBar and Options Menu Example

2. Develop an Android application demonstrating AlertDialog and custom DialogFragment.

ProjectName: DialogDemo

File Name: activity_main.xml




```
<?xml version="1.0" encoding="utf-8"?>

<LinearLayout

   xmlns:android="http://schemas.android.com/apk/res/android"

   android:id="@+id/main"

   android:layout_width="match_parent" android:layout_height="match_parent"

   android:orientation="vertical"
```

```
   android:gravity="center" android:padding="20dp">

   <Button

        android:id="@+id/btnAlert"

        android:layout_width="wrap_content"

        android:layout_height="wrap_content"

        android:text="Show AlertDialog"/>

   <Button

        android:id="@+id/btnCustom" android:layout_width="wrap_content"

        android:layout_height="wrap_content" android:text="Show Custom Dialog"

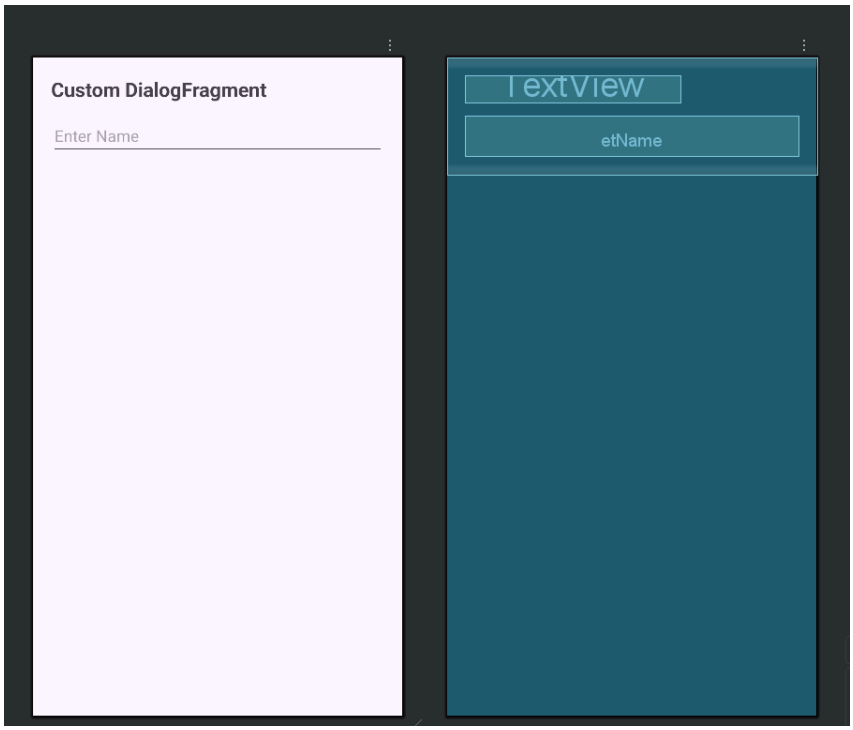
        android:layout_marginTop="20dp"/>

</LinearLayout>
```

Create Custom Dialog Layout.

Create:

res→ layout → New-> XML-> New Layout XML File → dialog_custom



```xml
<?xml version="1.0" encoding="utf-8"?>

<LinearLayout

   xmlns:android="http://schemas.android.com/apk/res/android"

   android:layout_width="match_parent"

   android:layout_height="wrap_content"

   android:orientation="vertical"

   android:padding="20dp">

   <TextView

       android:layout_width="wrap_content"

       android:layout_height="wrap_content"

       android:text="Custom DialogFragment"

       android:textSize="22sp"

       android:textStyle="bold"/>

   <EditText

       android:id="@+id/etName"

       android:layout_width="match_parent"

       android:layout_height="wrap_content"

       android:hint="Enter Name"

       android:layout_marginTop="15dp"/>

</LinearLayout>
```

Create Dialog Fragment Class.
Right Click: java package → New → Java Class

Class Name: CustomDialogFragment.java

```
package com.example.dialogdemo;



import android.app.AlertDialog;

import android.app.Dialog;

import android.os.Bundle;

import android.view.LayoutInflater;

import android.view.View;

import android.widget.EditText;

import android.widget.Toast;

import androidx.annotation.NonNull;

import androidx.fragment.app.DialogFragment;


public class CustomDialogFragment extends DialogFragment {


   EditText etName;

   @NonNull

   @Override

   public Dialog onCreateDialog(Bundle savedInstanceState) {


       AlertDialog.Builder builder = new AlertDialog.Builder(getActivity());

       LayoutInflater inflater = requireActivity().getLayoutInflater();

       View view = inflater.inflate(R.layout.dialog_custom, null);
```

```
        etName = view.findViewById(R.id.etName);

        builder.setView(view)

                .setTitle("Custom Dialog")

                .setPositiveButton("Submit", (dialog, which) -> {

                    String name = etName.getText().toString();

                    Toast.makeText(getActivity(),

                            "Entered: " + name,

                            Toast.LENGTH_SHORT).show();

                })

                .setNegativeButton("Cancel", (dialog, which) ->
dialog.dismiss());

        return builder.create();

    }

}
```

FileName: MainActivity.java

```
package com.example.dialogdemo;
```

```


import android.os.Bundle;

import android.widget.Button;

import android.widget.Toast;

import androidx.appcompat.app.AlertDialog;

import androidx.appcompat.app.AppCompatActivity;
```

```
public class MainActivity extends AppCompatActivity {



   Button btnAlert, btnCustom;

   @Override

   protected void onCreate(Bundle savedInstanceState) {

       super.onCreate(savedInstanceState);

       setContentView(R.layout.activity_main);



       btnAlert = findViewById(R.id.btnAlert);

       btnCustom = findViewById(R.id.btnCustom);



       // AlertDialog Example

       btnAlert.setOnClickListener(v -> {

           AlertDialog.Builder builder = new AlertDialog.Builder(this);

           builder.setTitle("Delete File")

                   .setMessage("Are you sure you want to delete?")

                   .setPositiveButton("YES", (dialog, which) -> {

                       Toast.makeText(this,

                               "File Deleted",

                               Toast.LENGTH_SHORT).show();



                   })

                   .setNegativeButton("NO", (dialog, which) ->
dialog.dismiss());
```

```
            builder.show();

        });

        // Custom Dialog Example

        btnCustom.setOnClickListener(v -> {

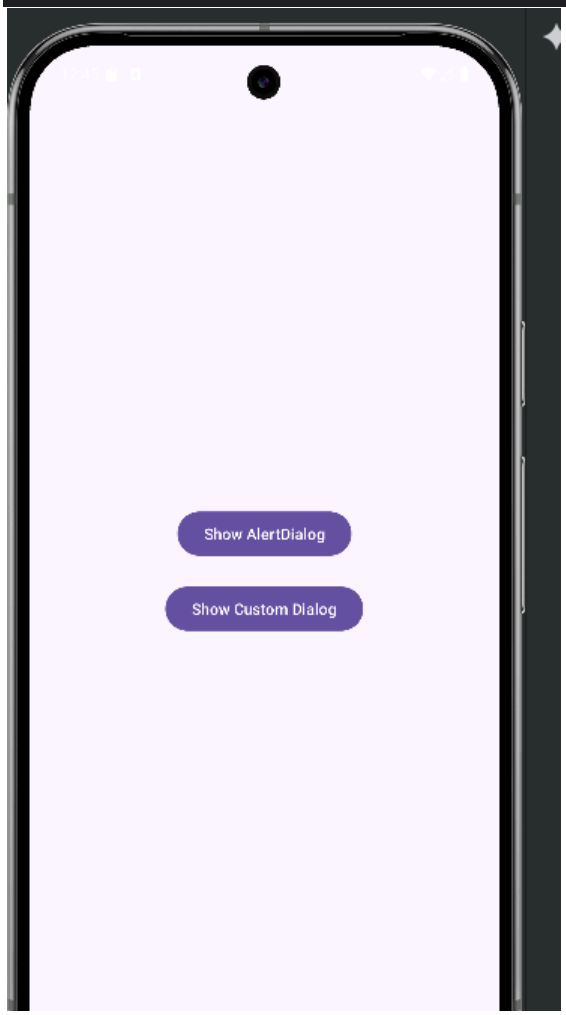
            CustomDialogFragment dialog = new CustomDialogFragment();

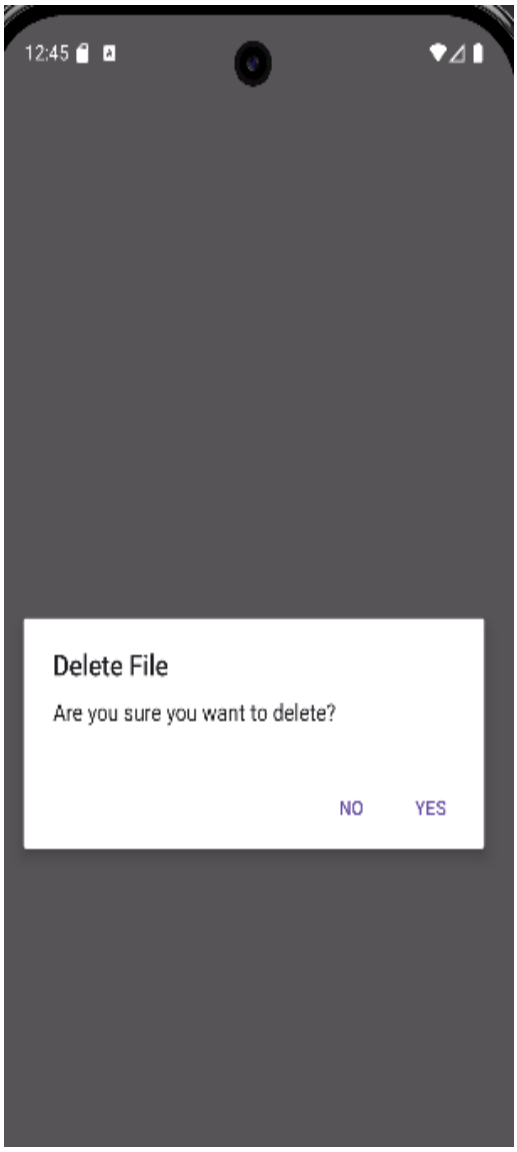
            dialog.show(getSupportFragmentManager(), "CustomDialog");

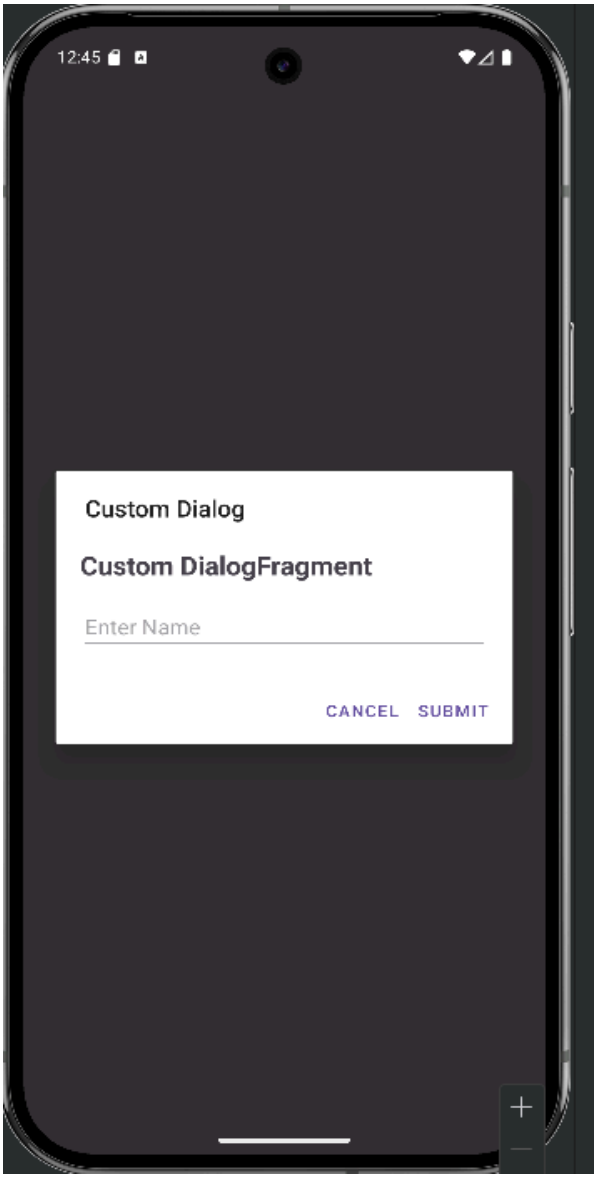
        });

    }

}
```




12:45
Delete File
Are you sure you want to delete?
NO
YES


12:45
Custom Dialog
Custom DialogFragment
Enter Name
CANCEL
SUBMIT

3. Develop an Android application implementing contextual menus and popup menus.

ProjectName: MenuDemo

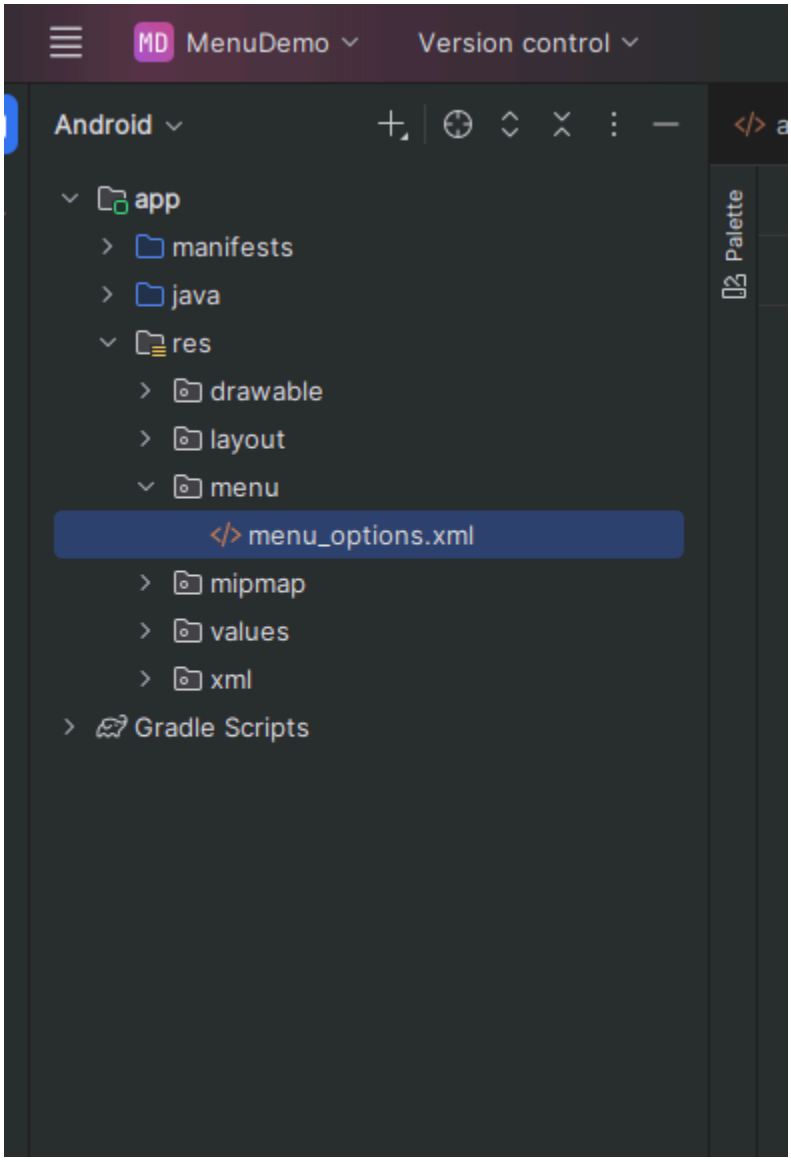
File Name: activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
```

```
<LinearLayout
   xmlns:android="http://schemas.android.com/apk/res/android"
   android:layout_width="match_parent"
   android:layout_height="match_parent"
   android:orientation="vertical"
   android:gravity="center"
   android:padding="20dp">
   <TextView
       android:id="@+id/txtContext"
       android:layout_width="wrap_content"
       android:layout_height="wrap_content"
       android:text="Long Press Me"
       android:textSize="24sp"
       android:padding="20dp"/>
   <Button
       android:id="@+id/btnPopup"
       android:layout_width="wrap_content"
       android:layout_height="wrap_content"
       android:text="Show Popup Menu"
       android:layout_marginTop="30dp"/>
</LinearLayout>
```

Create a Menu Resource File.
Right Click: res→ New → Android Resource Directory
Directory Name: menu
Resource Type: menu
Click OK
Right Click: menu → New → Menu Resource File

File Name : menu_options



File Name : menu_options.xml

```
<?xml version="1.0" encoding="utf-8"?>
```

```
<menu xmlns:android="http://schemas.android.com/apk/res/android">

   <item

       android:id="@+id/menu_edit" android:title="Edit"/>

   <item

       android:id="@+id/menu_delete"

       android:title="Delete"/>

   <item

       android:id="@+id/menu_share"

       android:title="Share"/>

</menu>
```

File Name: MainActivity.java

```
package com.example.menudemo;

import android.os.Bundle;

import android.view.ContextMenu;

import android.view.MenuItem;

import android.view.View;

import android.widget.Button;

import android.widget.PopupMenu;

import android.widget.TextView;

import android.widget.Toast;

import androidx.annotation.NonNull;

import androidx.appcompat.app.AppCompatActivity;
```

```java
public class MainActivity extends AppCompatActivity {


   TextView txtContext;

   Button btnPopup;


   @Override

   protected void onCreate(Bundle savedInstanceState) {

       super.onCreate(savedInstanceState);

       setContentView(R.layout.activity_main);


       txtContext = findViewById(R.id.txtContext);

       btnPopup = findViewById(R.id.btnPopup);


       // Register Context Menu

       registerForContextMenu(txtContext);


       // Popup Menu

       btnPopup.setOnClickListener(v -> {


           PopupMenu popupMenu = new PopupMenu(MainActivity.this, btnPopup);


           popupMenu.getMenuInflater().inflate(

                   R.menu.menu_options,

                   popupMenu.getMenu());
```

```
        popupMenu.setOnMenuItemClickListener(item -> {
            Toast.makeText(MainActivity.this,
                    item.getTitle(),
                    Toast.LENGTH_SHORT).show();
            return true;
        });

        popupMenu.show();
    });
}

// Context Menu Creation
@Override
public void onCreateContextMenu(ContextMenu menu,
                                View v,
                                ContextMenu.ContextMenuInfo menuInfo) {

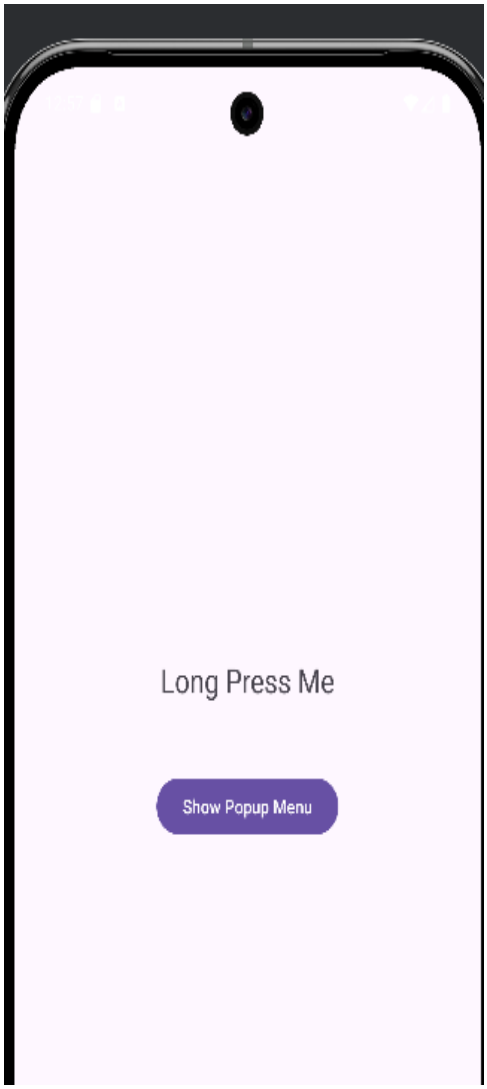
    super.onCreateContextMenu(menu, v, menuInfo);

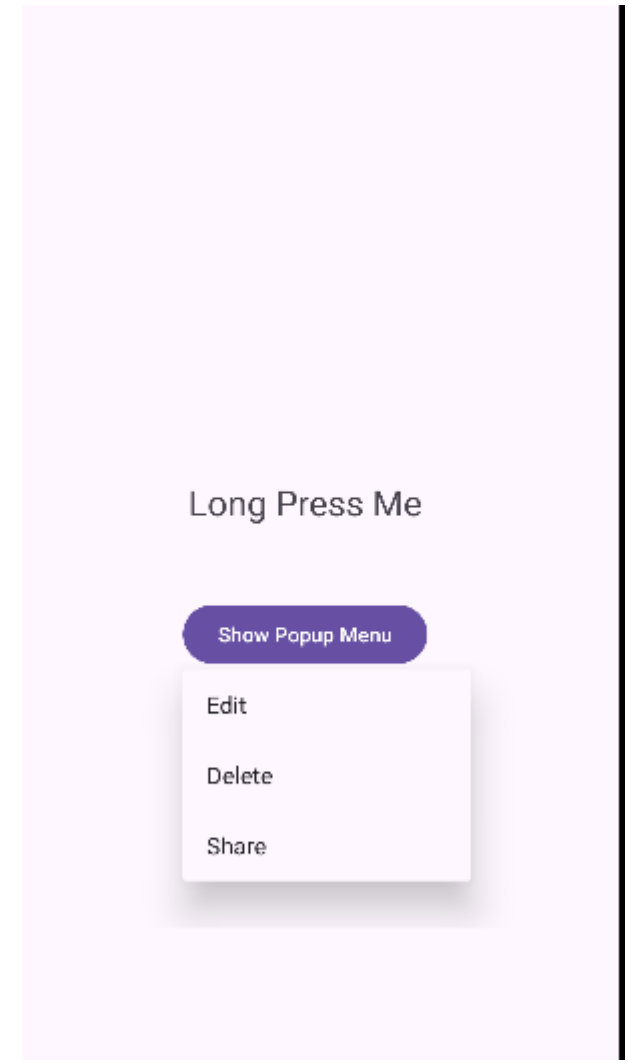
    menu.setHeaderTitle("Select Option");
    getMenuInflater().inflate(R.menu.menu_options, menu);
}

// Context Menu Click Handling
@Override
public boolean onContextItemSelected(@NonNull MenuItem item) {
```

```
        Toast.makeText(this,
                item.getTitle(),
                Toast.LENGTH_SHORT).show();


        return true;
    }
}
```





Intents, Events and Listeners

4. Develop an Android application implementing Explicit and Implicit Intents.

ProjectName: IntentDemo

File Name: activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout
   xmlns:android="http://schemas.android.com/apk/res/android"
   android:layout_width="match_parent" android:layout_height="match_parent"
   android:orientation="vertical"
   android:gravity="center"
   android:id="@+id/main"
   android:padding="20dp">
   <Button
       android:id="@+id/btnExplicit"
       android:layout_width="wrap_content"
       android:layout_height="wrap_content"
       android:text="Explicit Intent"/>
   <Button
       android:id="@+id/btnImplicit"
       android:layout_width="wrap_content"
       android:layout_height="wrap_content" android:text="Implicit Intent"
       android:layout_marginTop="20dp"/>
</LinearLayout>
```

Create Second Activity.

Right Click: java package → com.example.projectname -> New→ Activity → Empty Views Activity

Activity Name: SecondActivity
Click: Finish

Design Second Activity Layout.
Open: res→ layout → activity_second.xml

File Name: activity_second.xml

```
<?xml version="1.0" encoding="utf-8"?>
```

```
<LinearLayout

   xmlns:android="http://schemas.android.com/apk/res/android"

   android:layout_width="match_parent"

   android:layout_height="match_parent"

   android:id="@+id/main"

   android:gravity="center">

   <TextView

       android:layout_width="wrap_content"

       android:layout_height="wrap_content"

       android:text="Welcome to Second Activity"

       android:textSize="24sp"/>

</LinearLayout>
```

File Name: MainActivity.java.

```
package com.example.intentdemo;

import android.content.Intent;
import android.net.Uri;
import android.os.Bundle;
import android.widget.Button;
import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {

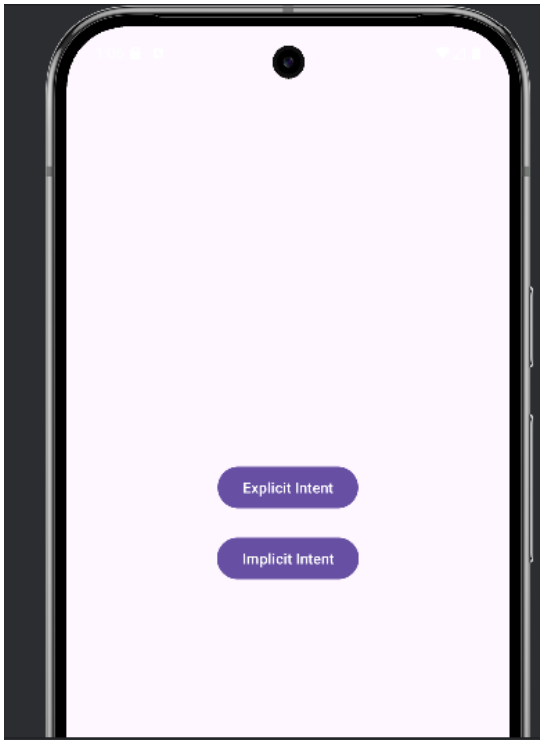
   Button btnExplicit, btnImplicit;

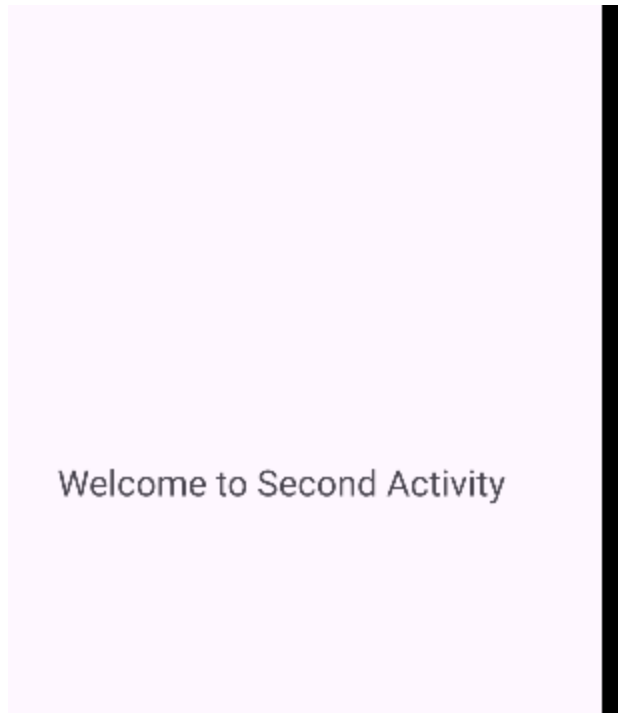
   @Override
   protected void onCreate(Bundle savedInstanceState) {
       super.onCreate(savedInstanceState);
       setContentView(R.layout.activity_main);

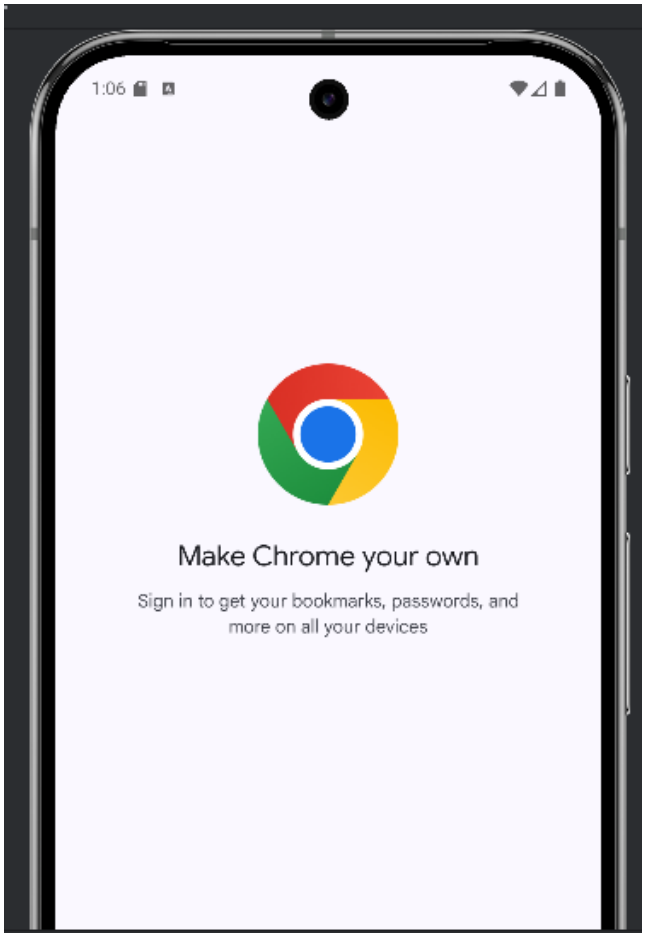
       btnExplicit = findViewById(R.id.btnExplicit);
       btnImplicit = findViewById(R.id.btnImplicit);

       // Explicit Intent
       btnExplicit.setOnClickListener(v -> {
           Intent intent = new Intent(MainActivity.this, SecondActivity.class);
           startActivity(intent);
       });
```

```
        // Implicit Intent
        btnImplicit.setOnClickListener(v -> {
            Intent intent = new Intent(
                    Intent.ACTION_VIEW,
                    Uri.parse("https://www.google.com")
            );
            startActivity(intent);
        });
    }
}
```


Explicit Intent
Implicit Intent


Welcome to Second Activity


1:06
Make Chrome your own
Sign in to get your bookmarks, passwords, and more on all your devices

5. Develop an Android application demonstrating Event Handling using Listeners (OnClick, OnLongClick, etc.).

Project Name: EventHandlingDemo

File Name: activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>

<LinearLayout

   xmlns:android="http://schemas.android.com/apk/res/android"

   android:layout_width="match_parent"

   android:layout_height="match_parent"

   android:orientation="vertical"

   android:gravity="center"

   android:padding="20dp">

   <Button

       android:id="@+id/btnClick"

       android:layout_width="wrap_content"

       android:layout_height="wrap_content"

       android:text="Click Me"/>

   <Button

       android:id="@+id/btnLongClick"

       android:layout_width="wrap_content"

       android:layout_height="wrap_content"

       android:text="Long Press Me"

       android:layout_marginTop="20dp"/>

</LinearLayout>
```

File Name: MainActivity.java

```
package com.example.eventhandlingdemo;


import android.os.Bundle;

import android.widget.Button;

import android.widget.Toast;

import androidx.appcompat.app.AppCompatActivity;


public class MainActivity extends AppCompatActivity {


   Button btnClick, btnLongClick;


   @Override

   protected void onCreate(Bundle savedInstanceState) {

       super.onCreate(savedInstanceState);

       setContentView(R.layout.activity_main);


       btnClick = findViewById(R.id.btnClick);

       btnLongClick = findViewById(R.id.btnLongClick);


       // OnClickListener

       btnClick.setOnClickListener(v -> {

           Toast.makeText(

                   MainActivity.this,

                   "Button Clicked",

                   Toast.LENGTH_SHORT
```

```
            ).show();

        });

        // OnLongClickListener

        btnLongClick.setOnLongClickListener(v -> {

            Toast.makeText(

                    MainActivity.this,

                    "Button Long Pressed",

                    Toast.LENGTH_SHORT

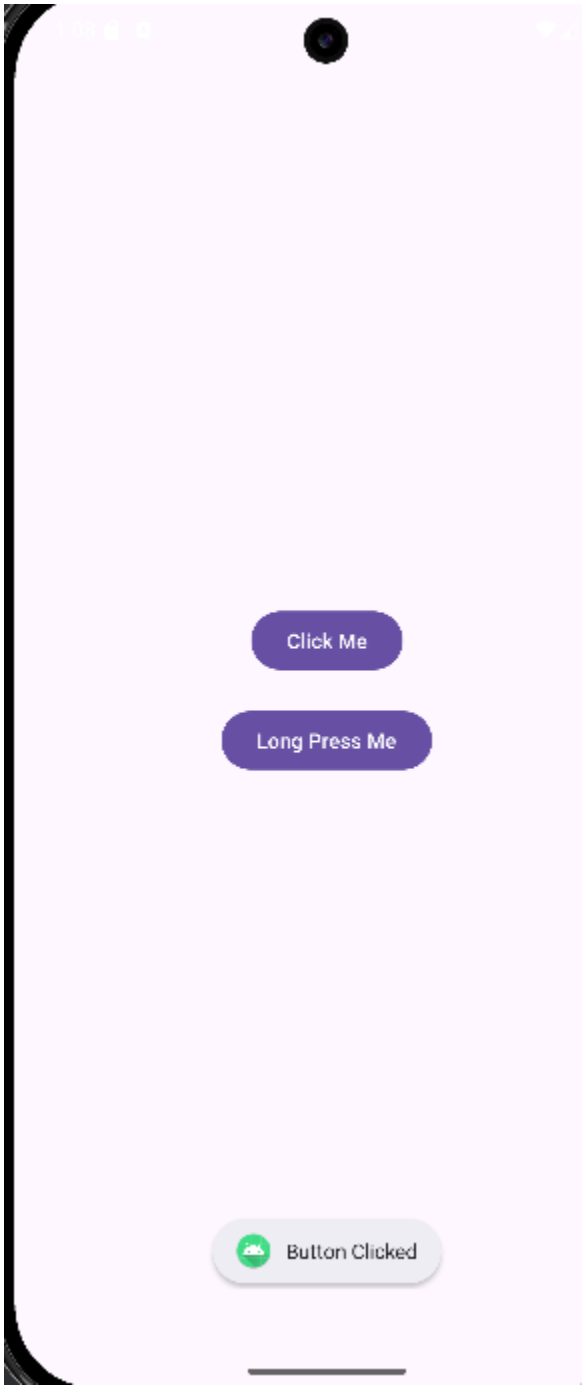
            ).show();

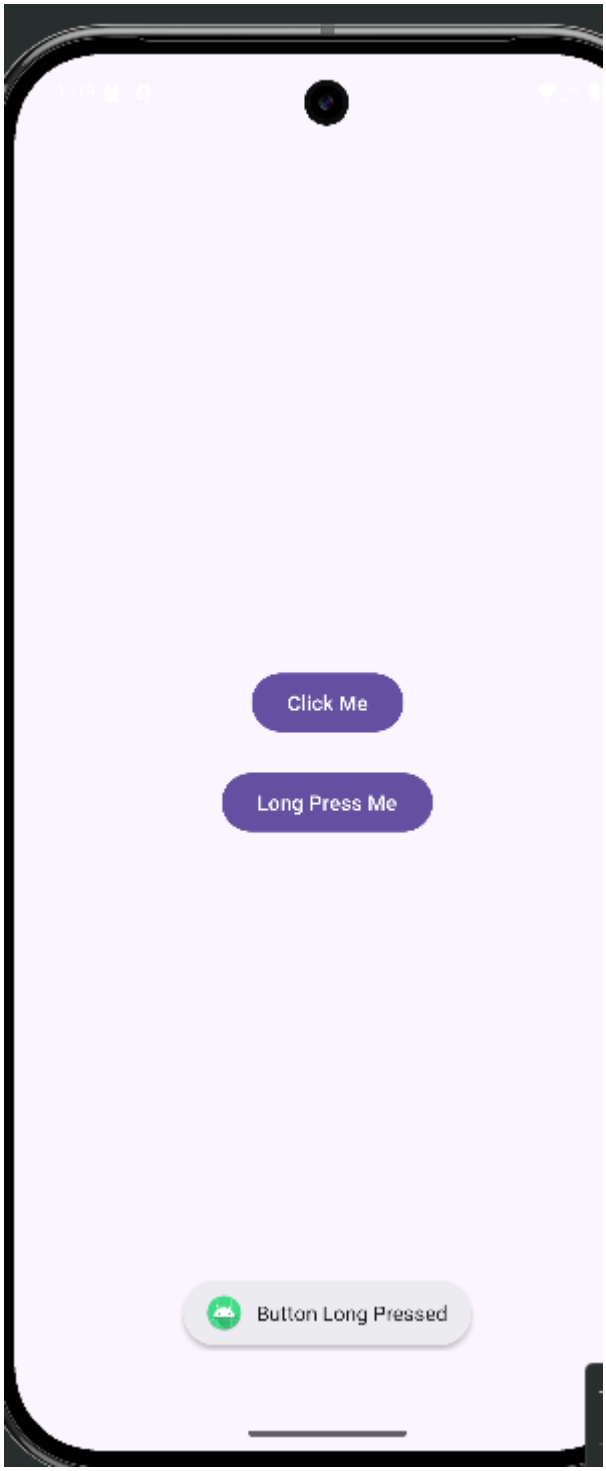
            return true;

        });

    }

}
```


Click Me
Long Press Me
Button Clicked


Click Me
Long Press Me
Button Long Pressed

6. Develop an Android application using Adapters to populate Spinner or ListView dynamically.

Android Spinner is a view similar to the dropdown list which is used to select one option from the list of options. It provides an easy way to select one item from the list of items and it shows a dropdown list of all values when we click on it. The default value of the android spinner will be the currently selected value and by using **Adapter** we can easily bind the items to the spinner objects. Generally, we populate our Spinner control with a list of items by using an **ArrayAdapter** in our Kotlin/Java file.

Project Name: AdapterDemo

File Name: activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>

<LinearLayout

   xmlns:android="http://schemas.android.com/apk/res/android"

   android:layout_width="match_parent"

   android:layout_height="match_parent"

   android:orientation="vertical"

   android:padding="20dp">

   <Spinner

       android:id="@+id/spinner"

       android:layout_width="match_parent"

       android:layout_height="wrap_content"/>

   <ListView

       android:id="@+id/listView"

       android:layout_width="match_parent"

       android:layout_height="wrap_content"

       android:layout_marginTop="20dp"/>

</LinearLayout>
```

File Name: MainActivity.java

```
package com.example.adapterdemo;


import android.os.Bundle;

import android.view.View;

import android.widget.AdapterView;

import android.widget.ArrayAdapter;

import android.widget.ListView;

import android.widget.Spinner;

import android.widget.Toast;

import androidx.appcompat.app.AppCompatActivity;


public class MainActivity extends AppCompatActivity {


   Spinner spinner;

   ListView listView;


   String[] cities = {

           "Mumbai",

           "Pune",

           "Delhi",

           "Chennai",

           "Hyderabad"

   };
```

```
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);


        spinner = findViewById(R.id.spinner);
        listView = findViewById(R.id.listView);

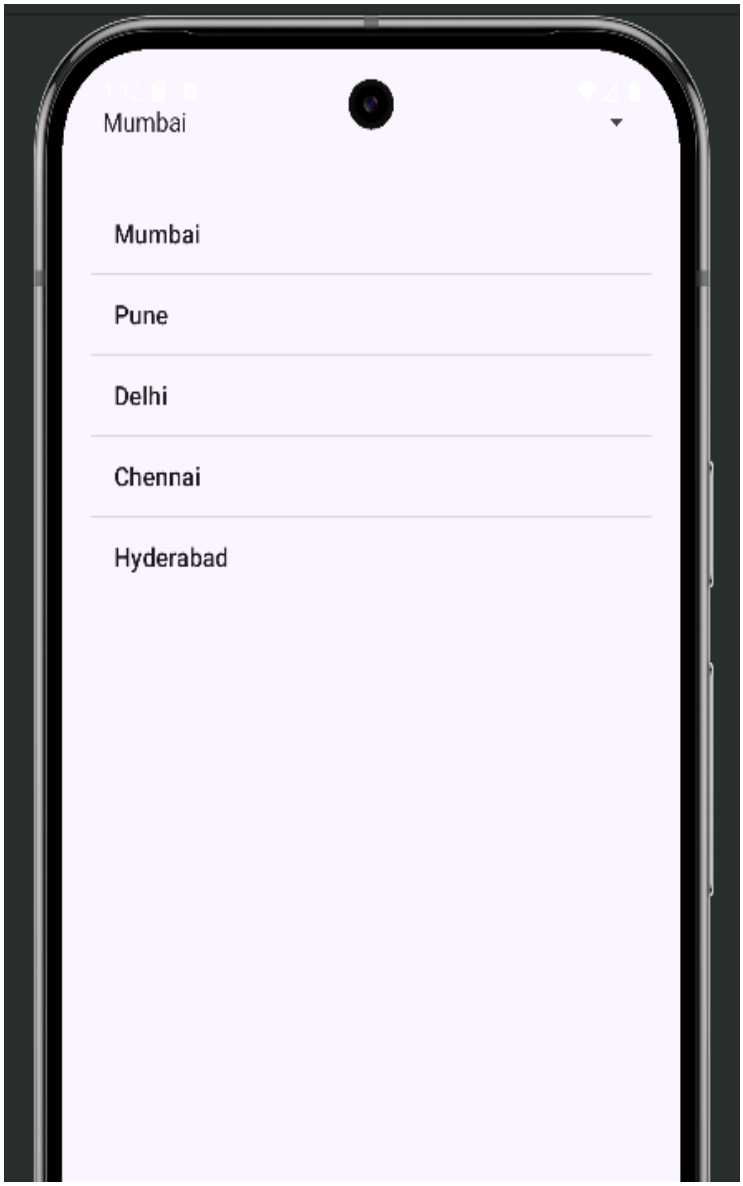

        // Spinner Adapter
        ArrayAdapter<String> spinnerAdapter = new ArrayAdapter<>(
                this,
                android.R.layout.simple_spinner_dropdown_item,
                cities
        );


        spinner.setAdapter(spinnerAdapter);
        // ListView Adapter
        ArrayAdapter<String> listAdapter = new ArrayAdapter<>(
                this,
                android.R.layout.simple_list_item_1,
                cities
        );
        listView.setAdapter(listAdapter);
        // Spinner Item Selection
        spinner.setOnItemSelectedListener(new
AdapterView.OnItemSelectedListener() {
```

```
            @Override
            public void onItemSelected(AdapterView<?> parent, View view,
                                       int position, long id) {


                Toast.makeText(
                        MainActivity.this,
                        "Selected: " + cities[position],
                        Toast.LENGTH_SHORT
                ).show();
            }
            @Override
            public void onNothingSelected(AdapterView<?> parent) {


            }
        });
        // ListView Item Click
        listView.setOnItemClickListener((parent, view, position, id) -> {
            Toast.makeText(
                    MainActivity.this,
                    "Clicked: " + cities[position],
                    Toast.LENGTH_SHORT
            ).show();
        });
    }
}
```


Mumbai
Mumbai
Pune
Delhi
Chennai
Hyderabad

Services, Notifications and Broadcast Receivers

7. Develop an Android application implementing Started and Bound Services.

A bound service is the server in a client-server interface. It lets components such as activities bind to the service, send requests, receive responses, and perform interprocess communication (IPC). A bound service typically lives only while it serves another application component and does not run in the background indefinitely.

Project Name: ServiceDemo

File Name: activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>

<LinearLayout

   xmlns:android="http://schemas.android.com/apk/res/android"

   android:layout_width="match_parent"

   android:layout_height="match_parent"

   android:orientation="vertical"

   android:gravity="center"

   android:padding="20dp">

   <Button

       android:id="@+id/btnStart"

       android:layout_width="wrap_content"

       android:layout_height="wrap_content"

       android:text="Start Service"/>

   <Button

       android:id="@+id/btnStop"

       android:layout_width="wrap_content"

       android:layout_height="wrap_content"
```

```
        android:text="Stop Service"

        android:layout_marginTop="15dp"/>

    <Button

        android:id="@+id/btnBound"

        android:layout_width="wrap_content"

        android:layout_height="wrap_content"

        android:text="Bind Service"

        android:layout_marginTop="15dp"/>

</LinearLayout>
```

Create a Started Service Class.

Right Click: java package→ New → Java Class

Class Name: MyStartedService

File Name: MyStartedService.java

```
package com.example.servicedemo;


import android.app.Service;

import android.content.Intent;

import android.os.IBinder;

import android.widget.Toast;


public class MyStartedService extends Service {


    @Override

    public int onStartCommand(Intent intent, int flags, int startId) {
```

```
        Toast.makeText(
                this,
                "Started Service Running",
                Toast.LENGTH_SHORT
        ).show();

        return START_STICKY;
    }

    @Override
    public void onDestroy() {
        super.onDestroy();

        Toast.makeText(
                this,
                "Started Service Stopped",
                Toast.LENGTH_SHORT
        ).show();
    }

    @Override
    public IBinder onBind(Intent intent) {
        return null;
    }
}
```

Create a Bound Service Class (Similar like above).

Create Java Class: MyBoundService

File Name:MyBoundService

```
package com.example.servicedemo;


import android.app.Service;

import android.content.Intent;

import android.os.Binder;

import android.os.IBinder;


public class MyBoundService extends Service {


   private final IBinder binder = new LocalBinder();


   public class LocalBinder extends Binder {

       public MyBoundService getService() {

           return MyBoundService.this;

       }

   }


   @Override

   public IBinder onBind(Intent intent) {

       return binder;

   }


   public String getMessage() {
```

```
        return "Bound Service Connected";
    }
}
```

File Name: MainActivity.java

```
package com.example.servicedemo;


import android.content.ComponentName;
import android.content.Intent;
import android.content.ServiceConnection;
import android.os.Bundle;
import android.os.IBinder;
import android.widget.Button;
import android.widget.Toast;


import androidx.appcompat.app.AppCompatActivity;


public class MainActivity extends AppCompatActivity {


    Button btnStart, btnStop, btnBound;


    MyBoundService myBoundService;
    boolean isBound = false;


    @Override
```

```
protected void onCreate(Bundle savedInstanceState) {

    super.onCreate(savedInstanceState);

    setContentView(R.layout.activity_main);



    btnStart = findViewById(R.id.btnStart);

    btnStop = findViewById(R.id.btnStop);

    btnBound = findViewById(R.id.btnBound);



    // Start Service

    btnStart.setOnClickListener(v -> {



        Intent intent = new Intent(

                this,

                MyStartedService.class

        );



        startService(intent);



    });



    // Stop Service

    btnStop.setOnClickListener(v -> {



        Intent intent = new Intent(

                this,
```

```
                MyStartedService.class
        );


        stopService(intent);


    });
    // Bind Service
    btnBound.setOnClickListener(v -> {
        Intent intent = new Intent(
                this,
                MyBoundService.class
        );


        bindService(
                intent,
                serviceConnection,
                BIND_AUTO_CREATE
        );
    });
}
// Service Connection
ServiceConnection serviceConnection = new ServiceConnection() {


    @Override
    public void onServiceConnected(
            ComponentName name,
```

```
                IBinder service) {

            MyBoundService.LocalBinder binder =

                    (MyBoundService.LocalBinder) service;

            myBoundService = binder.getService();

            isBound = true;

            Toast.makeText(

                    MainActivity.this,

                    myBoundService.getMessage(),

                    Toast.LENGTH_SHORT

            ).show();

        }



        @Override

        public void onServiceDisconnected(ComponentName name) {

            isBound = false;

        }

    };

    @Override

    protected void onDestroy() {

        super.onDestroy();

        if (isBound) {

            unbindService(serviceConnection);

            isBound = false;

        }

    }

}
```

Add Services in AndroidManifest.xml (in Android Project Panel).

Open:

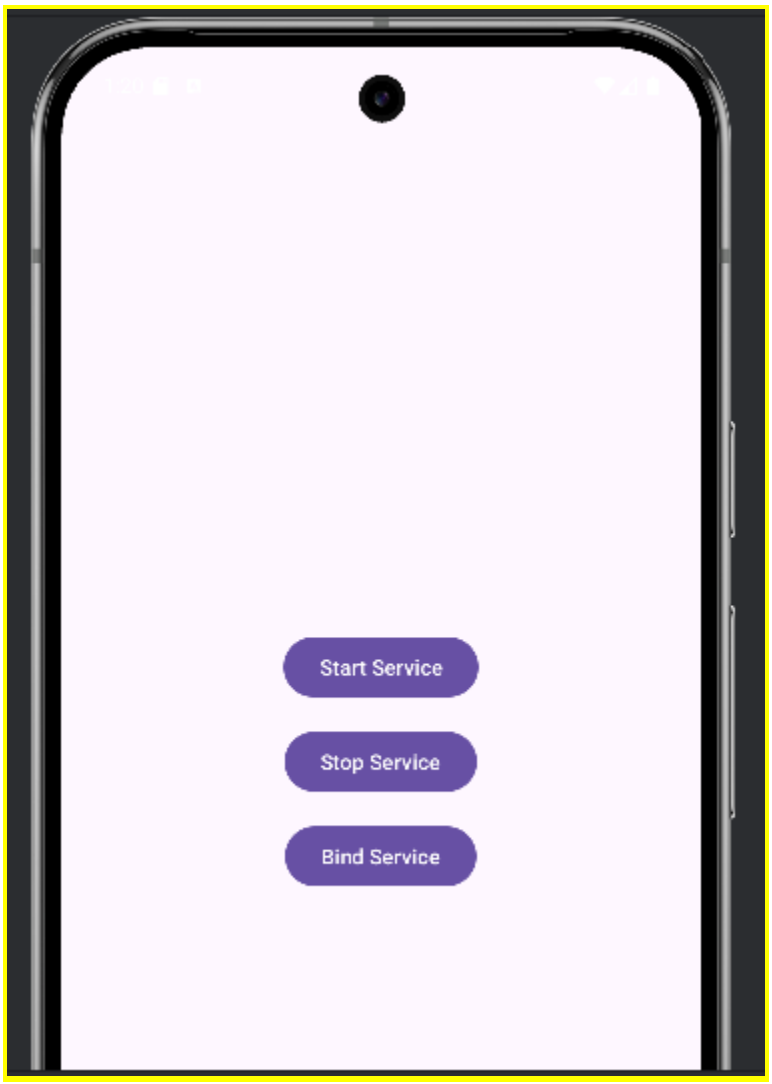
app→ Manifests → AndroidManifest.xml

Inside: <application>

Add:->

<service android:name=".MyStartedService"/>

<service android:name=".MyBoundService"/>



8. Develop an Android application implementing Notifications using Notification Manager.

Project Name: NotificationDemo

File Name: activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>

<LinearLayout

   xmlns:android="http://schemas.android.com/apk/res/android"

   android:layout_width="match_parent"

   android:layout_height="match_parent"

   android:gravity="center">

   <Button

       android:id="@+id/btnNotify"

       android:layout_width="wrap_content"

       android:layout_height="wrap_content"

       android:text="Show Notification"/>

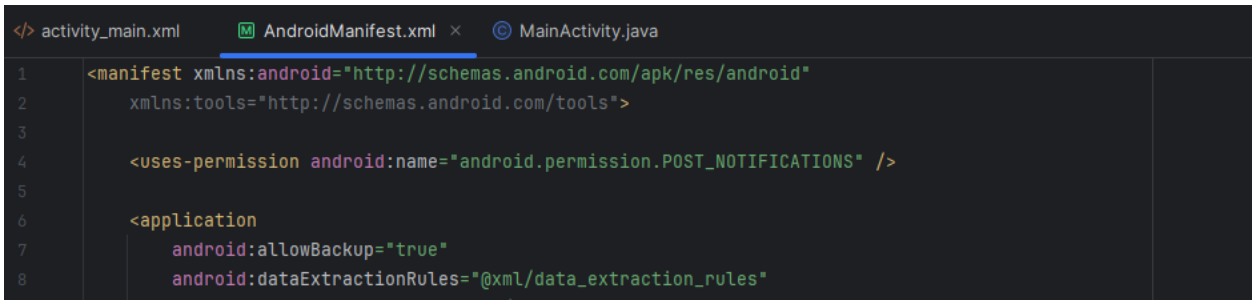
</LinearLayout>
```

Add Permission in Manifest,

Open: AndroidManifest.xml

Add above <application>:

<uses-permission android:name="android.permission.POST_NOTIFICATIONS”/>



File Name: MainActivity.java

```
package com.example.notificationdemo;


import android.Manifest;

import android.app.NotificationChannel;

import android.app.NotificationManager;

import android.content.pm.PackageManager;

import android.os.Build;

import android.os.Bundle;

import android.widget.Button;

import androidx.appcompat.app.AppCompatActivity;

import androidx.core.app.NotificationCompat;


public class MainActivity extends AppCompatActivity {


   private static final String CHANNEL_ID = "MyChannel";

   private static final int NOTIFICATION_PERMISSION_CODE = 100;


   private Button btnNotify;


   @Override

   protected void onCreate(Bundle savedInstanceState) {

       super.onCreate(savedInstanceState);

       setContentView(R.layout.activity_main);
```

```
        // Find the button from activity_main.xml
        btnNotify = findViewById(R.id.btnNotify);

        // Create notification channel
        createNotificationChannel();

        // Request notification permission for Android 13+
        requestNotificationPermission();

        // Button click
        btnNotify.setOnClickListener(v -> {
            showNotification();
        });
    }

    // Request notification permission
    private void requestNotificationPermission() {

        if (Build.VERSION.SDK_INT >= Build.VERSION_CODES.TIRAMISU) {

            if (checkSelfPermission(Manifest.permission.POST_NOTIFICATIONS)
                    != PackageManager.PERMISSION_GRANTED) {

                requestPermissions(
                        new String[]{Manifest.permission.POST_NOTIFICATIONS},
                        NOTIFICATION_PERMISSION_CODE
```

```
            );
        }
    }
}

// Create notification channel
private void createNotificationChannel() {

    if (Build.VERSION.SDK_INT >= Build.VERSION_CODES.O) {

        NotificationChannel channel = new NotificationChannel(
                CHANNEL_ID,
                "My Channel",
                NotificationManager.IMPORTANCE_DEFAULT
        );

        channel.setDescription("Notification Channel for Demo");

        NotificationManager manager =
                getSystemService(NotificationManager.class);

        if (manager != null) {
            manager.createNotificationChannel(channel);
        }
    }
}
```

```
// Show notification
private void showNotification() {

    // Check permission for Android 13+
    if (Build.VERSION.SDK_INT >= Build.VERSION_CODES.TIRAMISU) {

        if (checkSelfPermission(Manifest.permission.POST_NOTIFICATIONS)
                != PackageManager.PERMISSION_GRANTED) {

            requestNotificationPermission();
            return;
        }
    }

    NotificationCompat.Builder builder =
            new NotificationCompat.Builder(this, CHANNEL_ID)
                    .setSmallIcon(android.R.drawable.ic_dialog_info)
                    .setContentTitle("Notification Example")
                    .setContentText("Hello from Android Notification")
                    .setPriority(NotificationCompat.PRIORITY_DEFAULT)
                    .setAutoCancel(true);

    NotificationManager manager =
            (NotificationManager) getSystemService(NOTIFICATION_SERVICE);
```

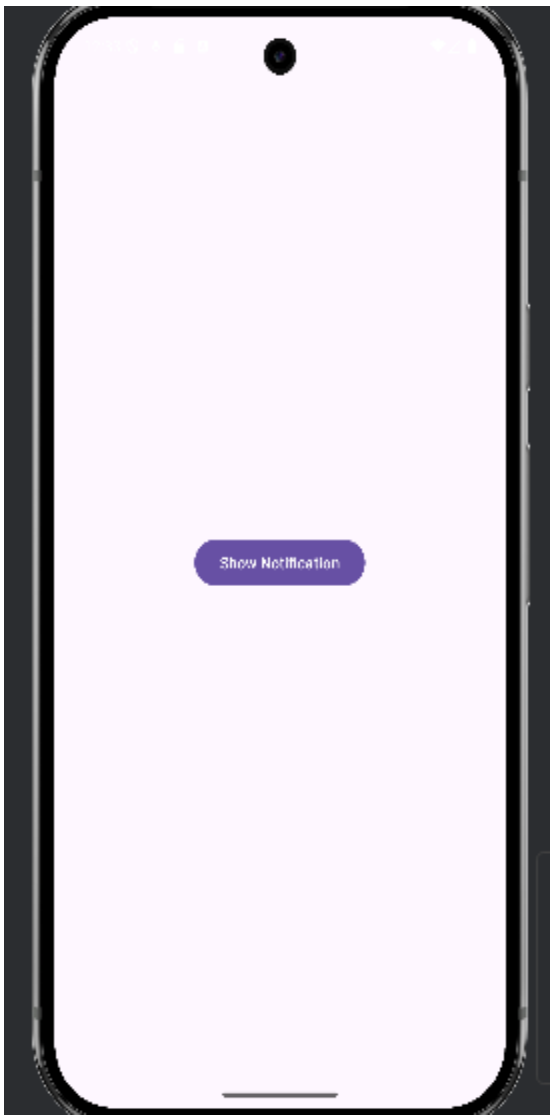
```
        if (manager != null) {

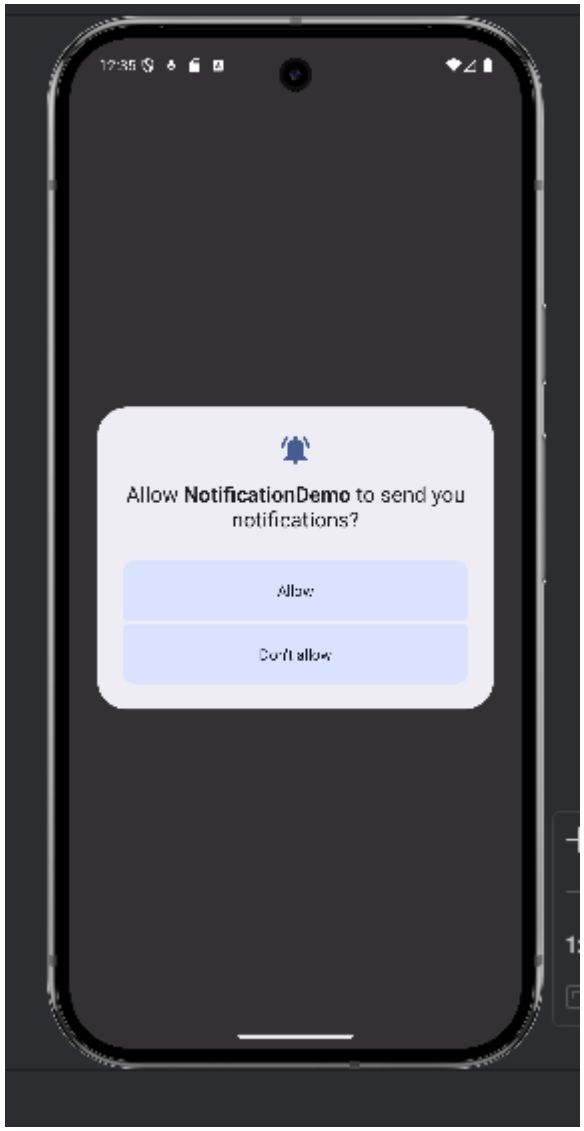
            manager.notify(1, builder.build());

        }

    }

}
```





9. Develop an Android application implementing Static and Dynamic Broadcast Receivers.

Project Name: BroadcastDemo

File Name: activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>

<LinearLayout

   xmlns:android="http://schemas.android.com/apk/res/android"

   android:layout_width="match_parent"

   android:layout_height="match_parent"

   android:gravity="center">

   <TextView

       android:layout_width="wrap_content"

       android:layout_height="wrap_content"

       android:text="Broadcast Receiver Example"

       android:textSize="22sp"/>

</LinearLayout>
```

Create a Broadcast Receiver Class.

Right Click:-> java package → New → Java Class Class Name: MyReceiver

```
package com.example.broadcastdemo;

import android.content.BroadcastReceiver;

import android.content.Context;

import android.content.Intent;

import android.widget.Toast;


public class MyReceiver extends BroadcastReceiver {


   @Override

   public void onReceive(Context context, Intent intent) {

       Toast.makeText(

               context,

               "Broadcast Received",

               Toast.LENGTH_SHORT

       ).show();

   }

}
```

Static Broadcast Receiver.

Open:AndroidManifest.xml

Inside: <application>

Add:

```
<receiver
   android:name=".MyReceiver"
   android:exported="true">
   <intent-filter>
       <action android:name="android.intent.action.AIRPLANE_MODE" />
   </intent-filter>
</receiver>
```

```
        android:supportsRtl="true"
        android:theme="@style/Theme.BroadcastDemo">

        <receiver
            android:name=".MyReceiver"
            android:exported="true">
            <intent-filter>
                <action android:name="android.intent.action.AIRPLANE_MODE" />
            </intent-filter>
        </receiver>

        <activity
            android:name=".MainActivity"
            android:exported="true">
            <intent-filter>
```

Dynamic Broadcast Receiver.

File Name: MainActivity.java

```
package com.example.broadcastdemo;

import android.content.IntentFilter;

import android.os.Bundle;

import androidx.appcompat.app.AppCompatActivity;

import androidx.core.content.ContextCompat;


public class MainActivity extends AppCompatActivity {


   MyReceiver myReceiver = new MyReceiver();


   @Override

   protected void onCreate(Bundle savedInstanceState) {

       super.onCreate(savedInstanceState);

       setContentView(R.layout.activity_main);


       // Dynamic Receiver

       IntentFilter filter = new IntentFilter();

       filter.addAction("android.intent.action.AIRPLANE_MODE");


       ContextCompat.registerReceiver(

               this,

               myReceiver,

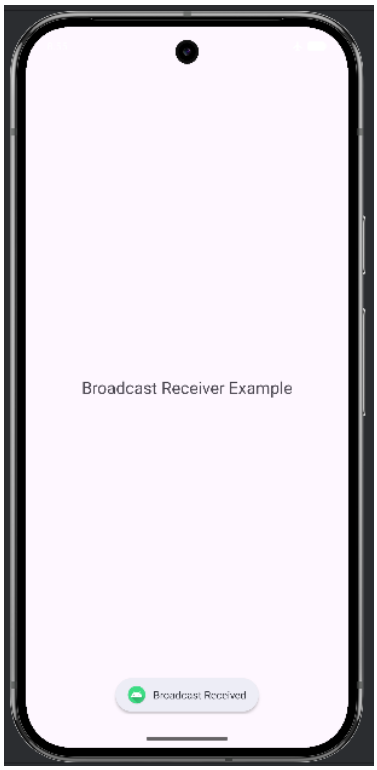
               filter,

               ContextCompat.RECEIVER_EXPORTED
```

```
        );
    }

    @Override
    protected void onDestroy() {
        super.onDestroy();
        unregisterReceiver(myReceiver);
    }
}
```

Turn on airplane mode



Database and Networking

10. Develop an Android application implementing SQLite database CRUD operations.

ProjectName: SQLLiteCRUDDemo

File Name: activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>

<LinearLayout

   xmlns:android="http://schemas.android.com/apk/res/android"

   android:layout_width="match_parent"

   android:layout_height="match_parent"

   android:orientation="vertical"

   android:padding="20dp">

   <EditText

       android:id="@+id/etName"

       android:layout_width="match_parent"

       android:layout_height="wrap_content"

       android:hint="Enter Name"/>

   <EditText

       android:id="@+id/etNewName"

       android:layout_width="match_parent"

       android:layout_height="wrap_content"

       android:hint="Enter New Name"/>

   <Button

       android:id="@+id/btnInsert"

       android:layout_width="match_parent"

       android:layout_height="wrap_content"
```

```
        android:text="Insert"/>

    <Button

        android:id="@+id/btnView"

        android:layout_width="match_parent"

        android:layout_height="wrap_content"

        android:text="View Data"/>

    <Button

        android:id="@+id/btnUpdate"

        android:layout_width="match_parent"

        android:layout_height="wrap_content"

        android:text="Update"/>

    <Button

        android:id="@+id/btnDelete"

        android:layout_width="match_parent"

        android:layout_height="wrap_content"

        android:text="Delete"/>

    <TextView

        android:id="@+id/txtResult" android:layout_width="match_parent"

        android:layout_height="wrap_content"

        android:textSize="18sp"

         android:paddingTop="20dp"/>

</LinearLayout>
```

Create a Database Helper Class.

Right Click:

java package → New → Java Class

Class Name: DBHelper

```
package com.example.sqllitecruddemo;


import android.content.ContentValues;

import android.content.Context;

import android.database.Cursor;

import android.database.sqlite.SQLiteDatabase;

import android.database.sqlite.SQLiteOpenHelper;


public class DBHelper extends SQLiteOpenHelper {


    public DBHelper(Context context) {

        super(context, "StudentDB", null, 1);

    }


    @Override

    public void onCreate(SQLiteDatabase db) {

        db.execSQL("CREATE TABLE student(name TEXT)");

    }


    @Override

    public void onUpgrade(SQLiteDatabase db, int oldVersion, int newVersion) {

        db.execSQL("DROP TABLE IF EXISTS student");
```

```
        onCreate(db);
    }


    // Insert
    public void insertData(String name) {
        SQLiteDatabase db = this.getWritableDatabase();


        ContentValues values = new ContentValues();
        values.put("name", name);


        db.insert("student", null, values);
    }


    // View
    public String viewData() {
        SQLiteDatabase db = this.getReadableDatabase();


        Cursor cursor = db.rawQuery("SELECT * FROM student", null);


        StringBuilder data = new StringBuilder();


        while (cursor.moveToNext()) {
            data.append(cursor.getString(0)).append("\n");
        }


        cursor.close();
```

```
    return data.toString();
}

// Update
public void updateData(String oldName, String newName) {
    SQLiteDatabase db = this.getWritableDatabase();

    ContentValues values = new ContentValues();
    values.put("name", newName);

    db.update(
            "student",
            values,
            "name=?",
            new String[]{oldName}
    );
}

// Delete
public void deleteData(String name) {
    SQLiteDatabase db = this.getWritableDatabase();

    db.delete(
            "student",
            "name=?",
```

```
                new String[]{name}

        );

    }

}
```

MainActivity.java

```
package com.example.sqllitecruddemo;



import android.os.Bundle;

import android.widget.Button;

import android.widget.EditText;

import android.widget.TextView;

import android.widget.Toast;



import androidx.appcompat.app.AppCompatActivity;



public class MainActivity extends AppCompatActivity {



    EditText etName, etNewName;

    TextView txtResult;



    Button btnInsert, btnView, btnUpdate, btnDelete;



    DBHelper dbHelper;
```

```java
@Override
protected void onCreate(Bundle savedInstanceState) {
    super.onCreate(savedInstanceState);
    setContentView(R.layout.activity_main);

    etName = findViewById(R.id.etName);
    etNewName = findViewById(R.id.etNewName);
    txtResult = findViewById(R.id.txtResult);

    btnInsert = findViewById(R.id.btnInsert);
    btnView = findViewById(R.id.btnView);
    btnUpdate = findViewById(R.id.btnUpdate);
    btnDelete = findViewById(R.id.btnDelete);

    dbHelper = new DBHelper(this);

    // Insert
    btnInsert.setOnClickListener(v -> {

        String name = etName.getText().toString();

        dbHelper.insertData(name);

        Toast.makeText(
                this,
                "Data Inserted",
```

```
                Toast.LENGTH_SHORT
        ).show();
    });

    // View
    btnView.setOnClickListener(v -> {

        String data = dbHelper.viewData();

        txtResult.setText(data);
    });

    // Update
    btnUpdate.setOnClickListener(v -> {

        String oldName = etName.getText().toString();
        String newName = etNewName.getText().toString();

        dbHelper.updateData(oldName, newName);

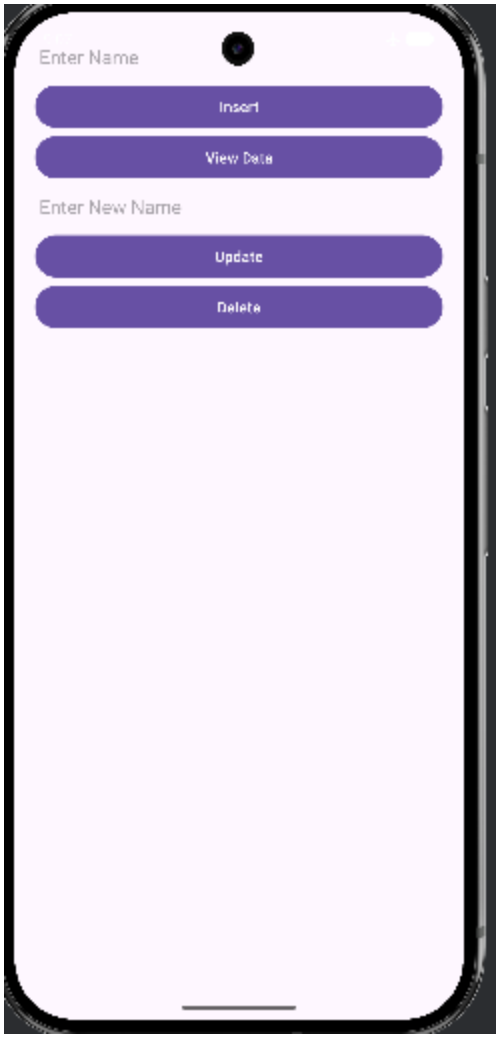
        Toast.makeText(
                this,
                "Data Updated",
                Toast.LENGTH_SHORT
        ).show();
    });
```

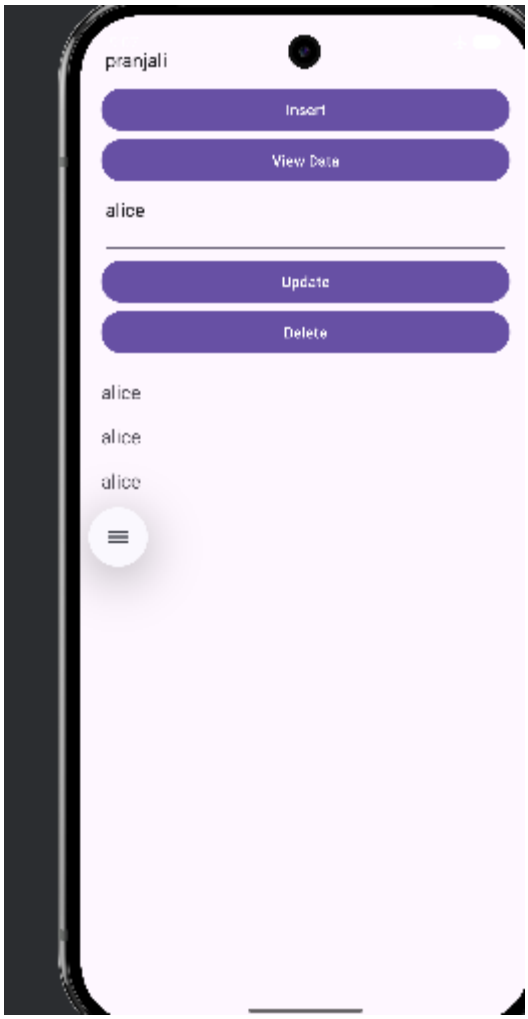
```
        // Delete
        btnDelete.setOnClickListener(v -> {

            String name = etName.getText().toString();

            dbHelper.deleteData(name);

            Toast.makeText(
                    this,
                    "Data Deleted",
                    Toast.LENGTH_SHORT
            ).show();
        });
    }
}
```


Enter Name
Insert
View Data
Enter New Name
Update
Delete


pranjali
Insert
View Data
alice
Update
Delete
alice
alice
alice

11. Develop an Android application integrating SQLite with ListView or RecyclerView.

Project Name : SQLiteCRUDDemo
activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout
xmlns:android="http://schemas.android.com/apk/res/android"
   android:layout_width="match_parent"
   android:layout_height="match_parent"
   android:orientation="vertical"
   android:padding="20dp">

   <EditText
       android:id="@+id/etName"
       android:layout_width="match_parent"
       android:layout_height="wrap_content"
       android:hint="Enter Name" />

   <Button
       android:id="@+id/btnInsert"
       android:layout_width="match_parent"
       android:layout_height="wrap_content"
       android:text="Insert" />
```

```
    <Button
        android:id="@+id/btnView"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="View Data" />
    <ListView
        android:id="@+id/listView"
        android:layout_width="match_parent"
        android:layout_height="wrap_content" />
</LinearLayout>
```

Create New Java Class in Package Folder

Name as: DBHelper

```
package com.example.sqlitecruddemo;


import android.content.ContentValues;
import android.content.Context;
import android.database.Cursor;
import android.database.sqlite.SQLiteDatabase;
import android.database.sqlite.SQLiteOpenHelper;


import java.util.ArrayList;


public class DBHelper extends SQLiteOpenHelper {


    public DBHelper(Context context) {
```

```
        super(context, "StudentDB", null, 1);
    }

    @Override
    public void onCreate(SQLiteDatabase db) {
        db.execSQL("CREATE TABLE student(name TEXT)");
    }

    @Override
    public void onUpgrade(SQLiteDatabase db, int oldVersion, int newVersion) {
        // Database upgrade code goes here
    }

    // Insert Data
    public void insertData(String name) {
        SQLiteDatabase db = this.getWritableDatabase();

        ContentValues values = new ContentValues();
        values.put("name", name);

        db.insert("student", null, values);
    }

    // Get All Data
    public ArrayList<String> getData() {
```

```
        ArrayList<String> list = new ArrayList<>();


        SQLiteDatabase db = this.getReadableDatabase();


        Cursor cursor = db.rawQuery("SELECT name FROM student", null);


        while (cursor.moveToNext()) {

            list.add(cursor.getString(0));

        }


        cursor.close();


        return list;

    }

}
```

MainActivity.java

```
package com.example.sqlitecruddemo;


import android.os.Bundle;

import android.widget.ArrayAdapter;

import android.widget.Button;

import android.widget.EditText;

import android.widget.ListView;

import android.widget.Toast;
```

```java
import androidx.appcompat.app.AppCompatActivity;


import java.util.ArrayList;


public class MainActivity extends AppCompatActivity {


   EditText etName;
   Button btnInsert, btnView;
   ListView listView;


   DBHelper dbHelper;


   @Override
   protected void onCreate(Bundle savedInstanceState) {
       super.onCreate(savedInstanceState);


       setContentView(R.layout.activity_main);


       etName = findViewById(R.id.etName);
       btnInsert = findViewById(R.id.btnInsert);
       btnView = findViewById(R.id.btnView);
       listView = findViewById(R.id.listView);


       dbHelper = new DBHelper(this);
```

```
        // Insert Data
        btnInsert.setOnClickListener(v -> {

            String name = etName.getText().toString();

            dbHelper.insertData(name);

            Toast.makeText(
                    this,
                    "Data Inserted",
                    Toast.LENGTH_SHORT
            ).show();

            etName.setText("");
        });

        // View Data
        btnView.setOnClickListener(v -> {

            ArrayList<String> list = dbHelper.getData();

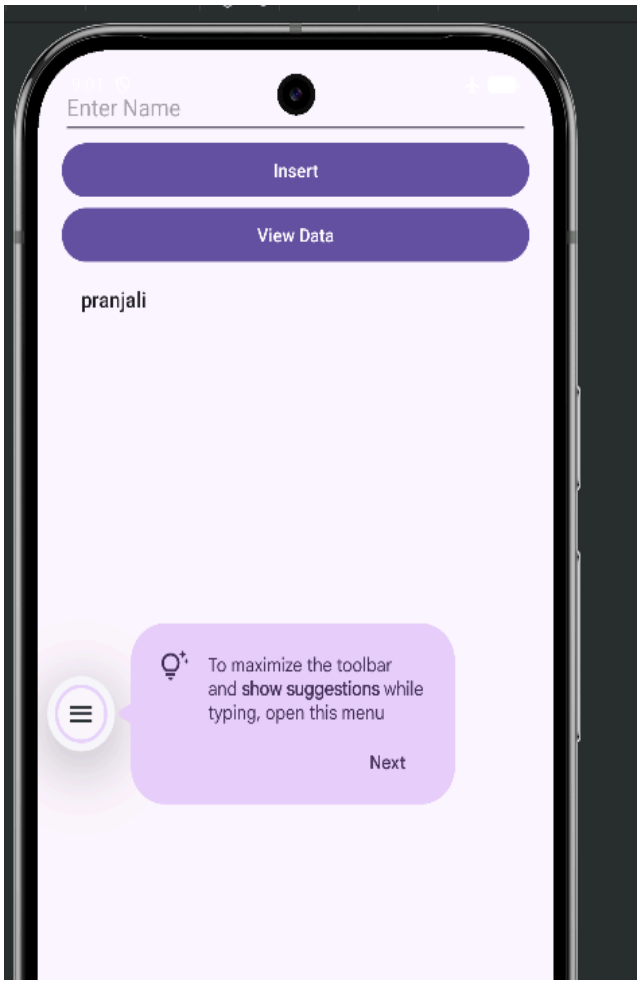
            ArrayAdapter<String> adapter = new ArrayAdapter<>(
                    this,
                    android.R.layout.simple_list_item_1,
                    list
            );
```

```
            listView.setAdapter(adapter);

        });

    }

}
```



12. Develop an Android application to fetch and parse JSON data from a web

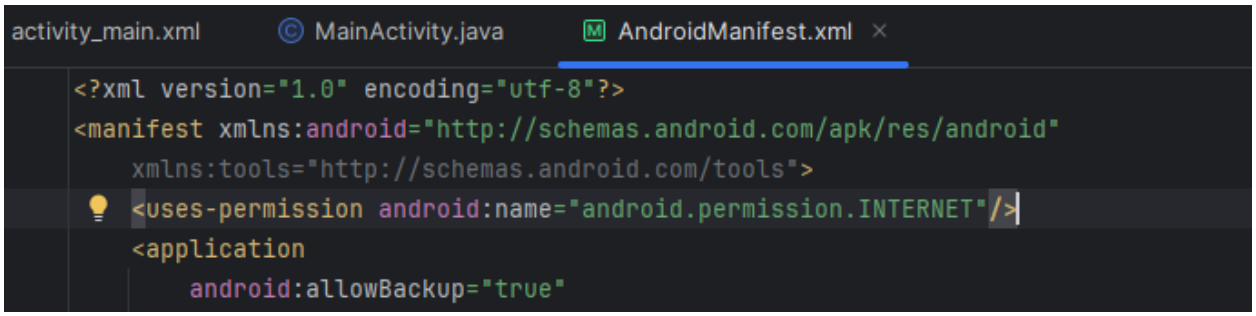
PROJECTNAME: JSONDEMO

Add Internet Permission.

AndroidManifest.xml

Add following above: <application> tag

<uses-permission android:name="android.permission.INTERNET"/>



activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
```

```
<LinearLayout

   xmlns:android="http://schemas.android.com/apk/res/android"

   android:layout_width="match_parent"

   android:layout_height="match_parent"

   android:orientation="vertical"

   android:padding="20dp">



   <Button

       android:id="@+id/btnFetch"

       android:layout_width="match_parent"

       android:layout_height="wrap_content"
```

```
        android:text="Fetch JSON Data" />


    <TextView
        android:id="@+id/txtResult"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:layout_marginTop="20dp"
        android:textSize="18sp"
        android:textColor="#000000" />


</LinearLayout>
```

MainActivity.java

package com.example.jsondemo;

```
package com.example.jsondemo;


import android.os.Bundle;
import android.widget.Button;
import android.widget.TextView;
import java.io.BufferedReader;
import java.io.InputStreamReader;
import java.net.HttpURLConnection;
import java.net.URL;
import org.json.JSONObject;
import androidx.appcompat.app.AppCompatActivity;
```

```
public class MainActivity extends AppCompatActivity {



    Button btnFetch;
    TextView txtResult;



    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);



        setContentView(R.layout.activity_main);



        btnFetch = findViewById(R.id.btnFetch);
        txtResult = findViewById(R.id.txtResult);



        btnFetch.setOnClickListener(v -> {
            fetchData();
        });
    }
```

```
   private void fetchData() {



       new Thread(() -> {



           try {



               // Web Service URL

               URL url = new
URL("https://jsonplaceholder.typicode.com/posts/1");



               HttpURLConnection connection =

                       (HttpURLConnection) url.openConnection();


               connection.setRequestMethod("GET");



               connection.setRequestMethod("GET");



               BufferedReader reader = new BufferedReader(new
InputStreamReader(

                               connection.getInputStream()
```

```
        )
);



StringBuilder result = new StringBuilder();



String line;



while ((line = reader.readLine()) != null) {
    result.append(line);
}



reader.close();



// JSON Parsing
JSONObject jsonObject =
        new JSONObject(result.toString());



String title =
        jsonObject.getString("title");
```

```
            String body =
                    jsonObject.getString("body");



            // Display Data
            runOnUiThread(() -> {



                txtResult.setText(
                        "Title: " + title +
                                "\n\nBody: " + body
                );
            });


            connection.disconnect();


        } catch (Exception e) {
            e.printStackTrace();
            runOnUiThread(() -> {
                txtResult.setText(
                        "Error: " + e.getMessage()
                );
            });
        }
```

```
        }).start();

    }

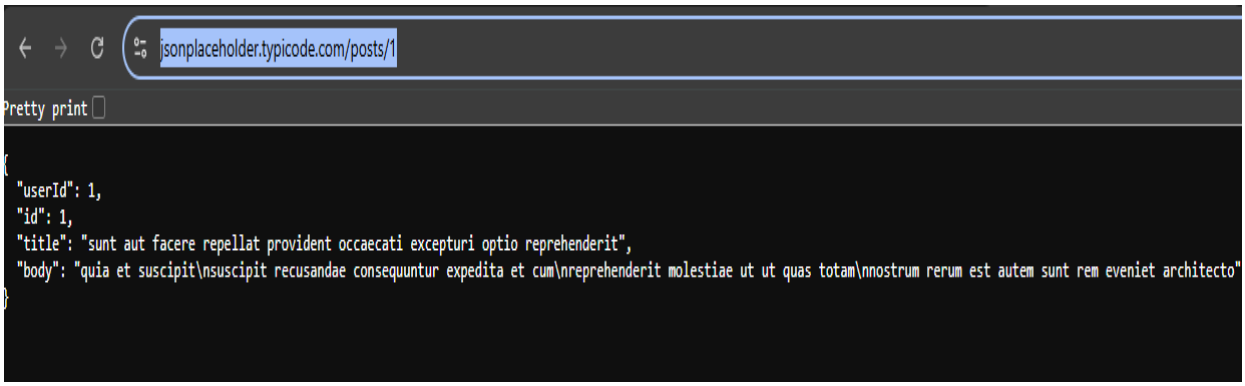
}
```

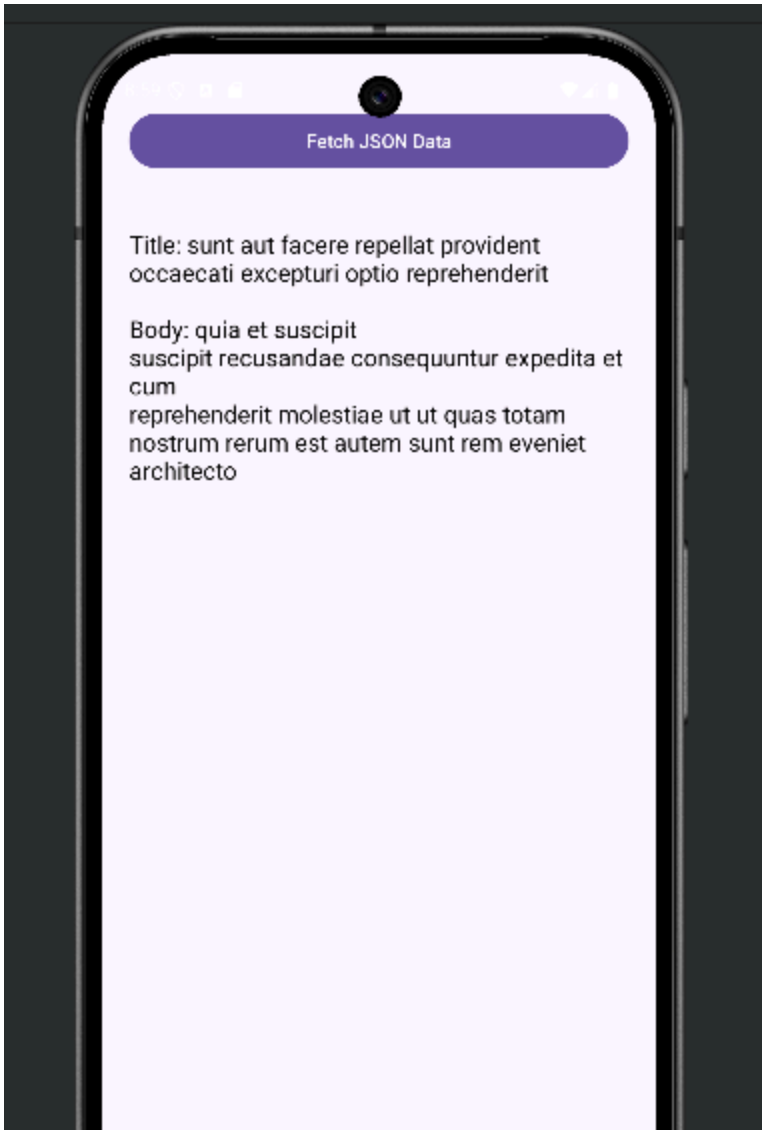
Step 6 : Web Service Used.

API URL : ***https://jsonplaceholder.typicode.com/posts/1***

This is a free test API for students.

https://jsonplaceholder.typicode.com/posts/1




Fetch JSON Data
Title: sunt aut facere repellat provident occaecati excepturi optio reprehenderit
Body: quia et suscipit
suscipit recusandae consequuntur expedita et cum
reprehenderit molestiae ut ut quas totam
nostrum rerum est autem sunt rem eveniet architecto

Concurrency, Media and Security

13. Develop an Android application demonstrating Threads and Handlers for background processing.

Project Name: ThreadHandlerDemo

Activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout
   xmlns:android="http://schemas.android.com/apk/res/android"
   android:layout_width="match_parent"
   android:layout_height="match_parent"
   android:orientation="vertical"
   android:gravity="center"
   android:padding="20dp">
   <Button
       android:id="@+id/btnStart"
       android:layout_width="wrap_content"
       android:layout_height="wrap_content"
       android:text="Start Background Task"/>
   <TextView
       android:id="@+id/txtResult"
       android:layout_width="wrap_content"
       android:layout_height="wrap_content"
       android:textSize="22sp"
       android:paddingTop="20dp"/></LinearLayout>
```

MainActivity.java

```
package com.example.threadhandlerdemo;

import android.os.Bundle;
import android.os.Handler;
import android.os.Looper;
import android.widget.Button;
import android.widget.TextView;

import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {

    Button btnStart;
    TextView txtResult;

    Handler handler = new Handler(Looper.getMainLooper());

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        btnStart = findViewById(R.id.btnStart);
        txtResult = findViewById(R.id.txtResult);
```

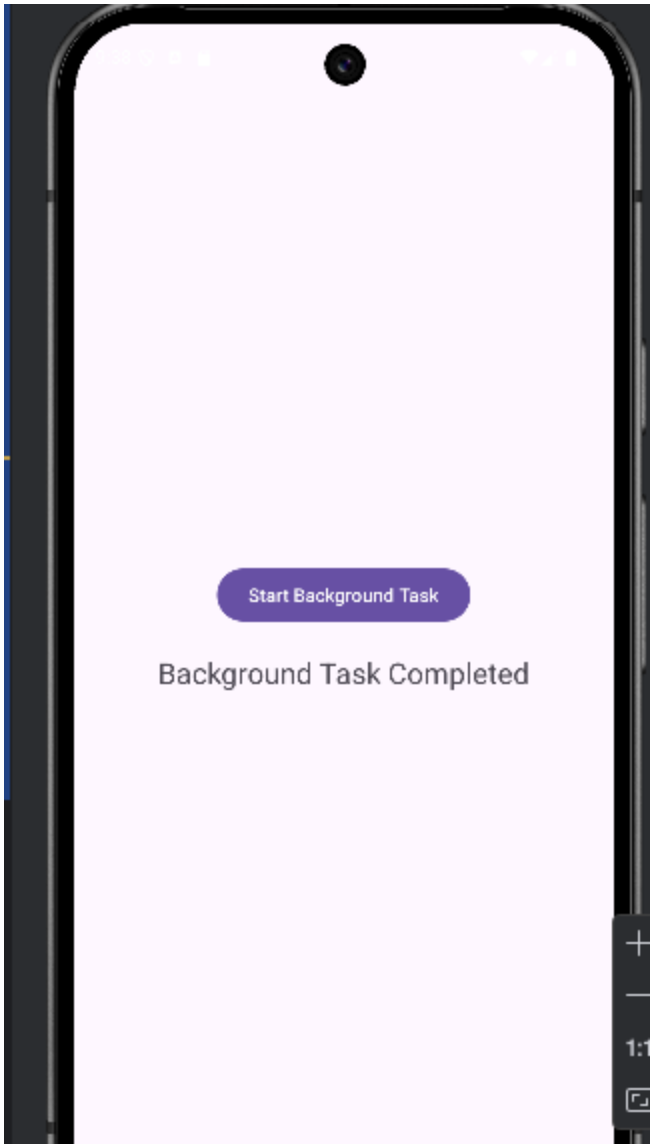
```
        btnStart.setOnClickListener(v -> {


            // Background Thread
            Thread thread = new Thread(() -> {
                try {
                    // Simulate Background Work
                    Thread.sleep(3000);


                } catch (Exception e) {
                    e.printStackTrace();
                }


                // Update UI using Handler
                handler.post(() -> {
                    txtResult.setText("Background Task Completed");
                });
            });


            thread.start();
        });
    }
}
```


Start Background Task
Background Task Completed

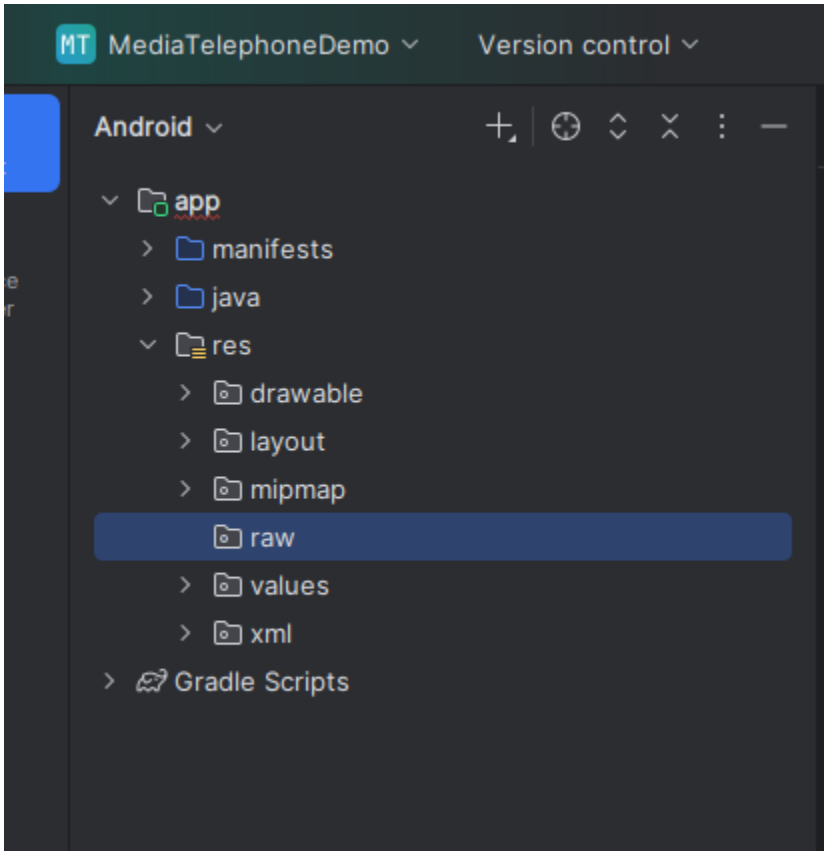
14. Develop an Android application implementing Media APIs and Telephone APIs.

Project Name: MediaTelephoneDemo

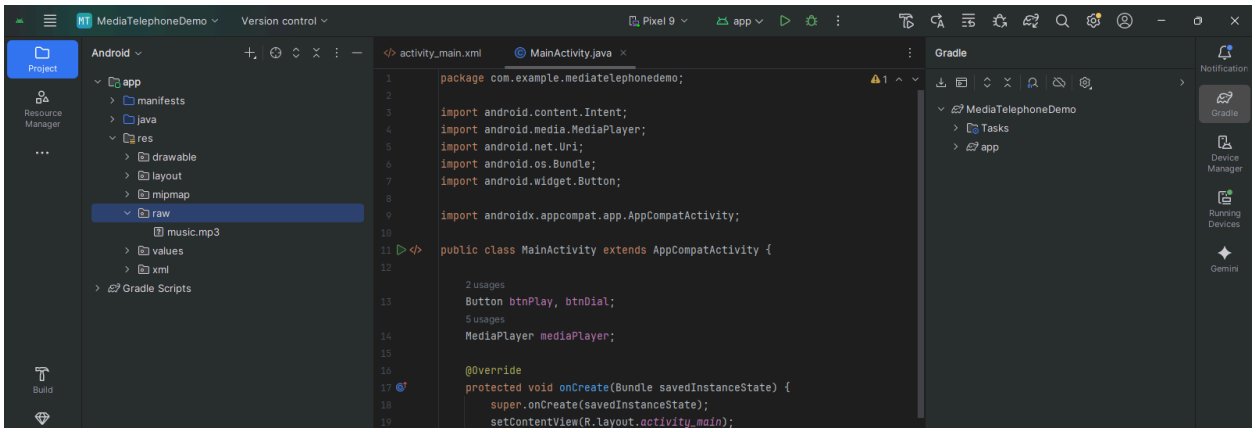
activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout
   xmlns:android="http://schemas.android.com/apk/res/android"
   android:layout_width="match_parent"
   android:layout_height="match_parent"
   android:orientation="vertical"
   android:gravity="center"
   android:padding="20dp">
   <Button
       android:id="@+id/btnPlay"
       android:layout_width="wrap_content"
       android:layout_height="wrap_content"
       android:text="Play Music"/>
   <Button
       android:id="@+id/btnDial"
       android:layout_width="wrap_content"
       android:layout_height="wrap_content"
       android:text="Open Dialer"
       android:layout_marginTop="20dp"/> </LinearLayout>
```

Add Audio File.
Create New folder in Android Project Panel: res raw



Download or use pre-existing small size MP3 file inside raw folder: music.mp3



MainActivity.java.

```
package com.example.mediatelephonedemo;
```

```
import android.content.Intent;
import android.media.MediaPlayer;
import android.net.Uri;
import android.os.Bundle;
import android.widget.Button;

import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {

   Button btnPlay, btnDial;
   MediaPlayer mediaPlayer;

   @Override
   protected void onCreate(Bundle savedInstanceState) {
       super.onCreate(savedInstanceState);
       setContentView(R.layout.activity_main);

       btnPlay = findViewById(R.id.btnPlay);
       btnDial = findViewById(R.id.btnDial);

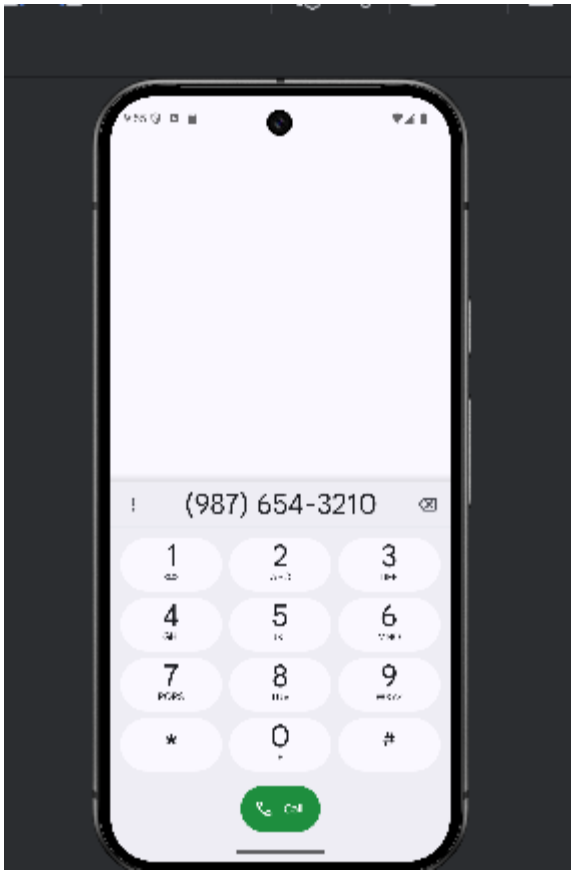
       // Media API
       mediaPlayer = MediaPlayer.create(this, R.raw.music);
```

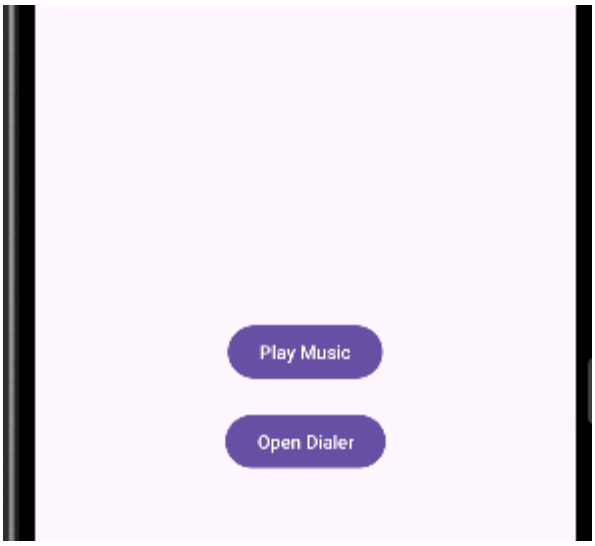
```
        // Play music
        btnPlay.setOnClickListener(v -> {
            mediaPlayer.start();
        });

        // Open phone dialer
        btnDial.setOnClickListener(v -> {
            Intent intent = new Intent(Intent.ACTION_DIAL);
            intent.setData(Uri.parse("tel:9876543210"));
            startActivity(intent);
        });
    }

    @Override
    protected void onDestroy() {
        super.onDestroy();

        if (mediaPlayer != null) {
            mediaPlayer.release();
            mediaPlayer = null;
        }
    }
}
```


(987) 654-3210
1
2
3
4
5
6
7
8
9
*
0
#


Play Music
Open Dialer

15. Develop an Android application implementing Runtime Permissions and security handling in AndroidManifest.xml

Project Name:PermissionDemo

Add Permission in AndroidManifest.xml.
Open:
AndroidManifest.xml
Add below code lines above: <application> tag
<uses-permission android:name="android.permission.CAMERA"/>
<uses-feature android:name="android.hardware.camera"
android:required="false" />

```
<manifest xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:tools="http://schemas.android.com/tools">
    <uses-permission android:name="android.permission.CAMERA"/>
    <uses-feature android:name="android.hardware.camera"
        android:required="false" />

    <application
        android:allowBackup="true"
        android:dataExtractionRules="@xml/data_extraction_rules"
        android:fullBackupContent="@xml/backup_rules"
        android:icon="@mipmap/ic_launcher"
```

activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:gravity="center">
    <Button
        android:id="@+id/btnPermission"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
```

```
        android:text="Request Camera Permission"/>

</LinearLayout>
```

MainActivity.java

```
package com.example.permissiondemo;


import android.content.pm.PackageManager;

import android.os.Bundle;

import android.widget.Button;

import android.widget.Toast;

import androidx.appcompat.app.AppCompatActivity;

import androidx.core.app.ActivityCompat;

import androidx.core.content.ContextCompat;


public class MainActivity extends AppCompatActivity {


    Button btnPermission;


    @Override

    protected void onCreate(Bundle savedInstanceState) {

        super.onCreate(savedInstanceState);

        setContentView(R.layout.activity_main);


        btnPermission = findViewById(R.id.btnPermission);
```

```
    btnPermission.setOnClickListener(v -> {
        checkPermission();
    });
}

private void checkPermission() {

    // Check if Camera permission is already granted
    if (ContextCompat.checkSelfPermission(
            this,
            android.Manifest.permission.CAMERA
    ) == PackageManager.PERMISSION_GRANTED) {

        Toast.makeText(
                this,
                "Permission Already Granted",
                Toast.LENGTH_SHORT
        ).show();

    } else {

        // Request Camera permission
        ActivityCompat.requestPermissions(
                this,
                new String[]{
                        android.Manifest.permission.CAMERA
```

```
                },
                1
        );
    }
}

// Permission Result
@Override
public void onRequestPermissionsResult(
        int requestCode,
        String[] permissions,
        int[] grantResults
) {
    super.onRequestPermissionsResult(
            requestCode,
            permissions,
            grantResults
    );

    if (requestCode == 1) {

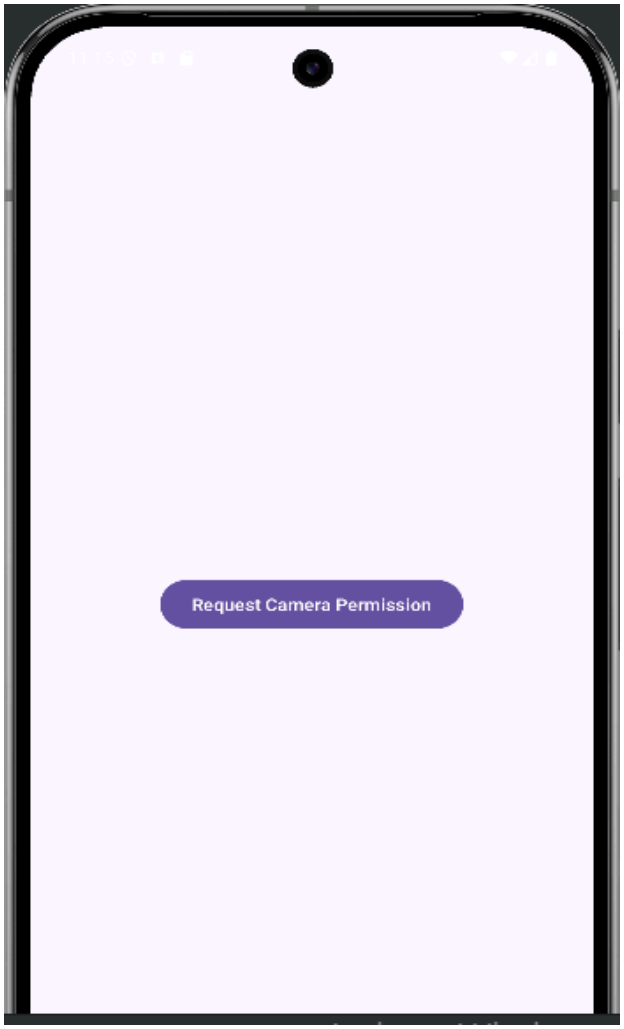
        if (grantResults.length > 0
                && grantResults[0] == PackageManager.PERMISSION_GRANTED) {

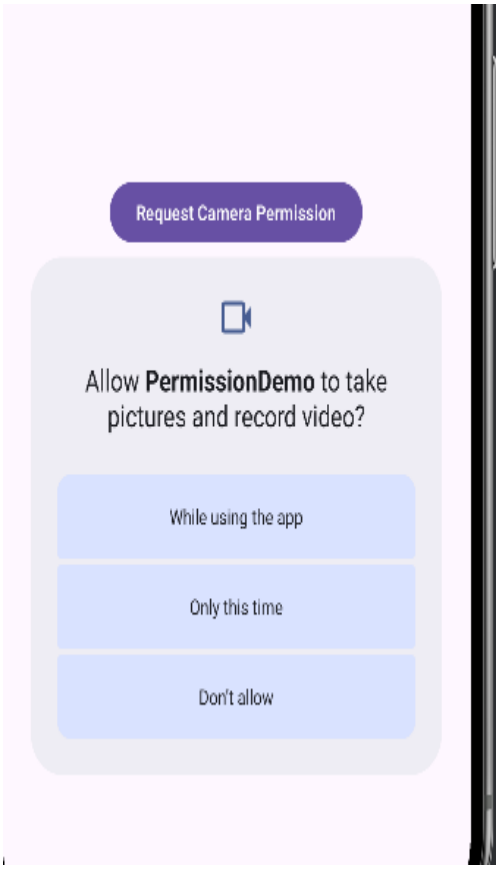
            Toast.makeText(
                    this,
```

```
                        "Permission Granted",
                        Toast.LENGTH_SHORT
                ).show();


            } else {


                Toast.makeText(
                        this,
                        "Permission Denied",
                        Toast.LENGTH_SHORT
                ).show();
            }
        }
    }
}
```


Request Camera Permission


Request Camera Permission
Allow PermissionDemo to take pictures and record video?
While using the app
Only this time
Don't allow